Does Youth Representation Matter for Social Spending?

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Abstract

Young people are underrepresented in most political institutions. While studies have investigated the causes behind the shortage of younger politicians, the potential consequences for the substantive representation of young people's interests in policy outcomes remains under-researched. I argue that younger politicians will be more likely than older politicians to allocate social spending toward child welfare for young families. Using an original dataset of over 12,000 mayoral candidates in Japan (2006–2019) and a regression discontinuity design, supplemented by interviews with 15 mayors, I find that municipalities that elect younger mayors increase spending on child welfare, especially through long-term investments in infrastructure. Mechanism tests and interviews suggest these effects stem from younger mayors' different life stages and longer time horizons rather than greater electoral incentives to court younger voters. Notably, younger mayors do not cut elderly welfare spending, suggesting youth representation can benefit young families without necessarily exacerbating intergenerational conflict over welfare.

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Young people are underrepresented in most political institutions. Over half of the world's eligible voters are under 40 years old, compared to 15% of national legislators (Inter-Parliamentary Union 2018). In many advanced democracies such as Japan, this disparity is even greater. Less than 5% of members of Japan's House of Representatives are under 40, compared to nearly a third of the voting-age population. Local politicians play a critical role in the provision of social services, yet they too tend to be older than most of their constituents. The average municipal assembly member begins their four-year term at 59 and the average mayor at 62. Just 6% of assembly members and 2% of mayors are under 40.[1]

Although scholars have studied the *causes* of the age bias in political institutions (Lawless and Fox 2015; Stockemer and Sundström 2022), there is a lack of research on the potential *consequences* for the substantive representation of young people's interests in policy outcomes. This is surprising given that there are well-developed literatures on how other characteristics such as a politician's race, gender, class, and sexual orientation can influence the extent to which they advocate on behalf of groups who share similar characteristics (e.g., Broockman 2013; Carnes 2012; Reynolds 2013; Schwindt-Bayer 2006). Moreover, the shortage of younger politicians across all levels of government may be concerning for several reasons. There are many issues that disproportionately affect the young, from policies on education, unemployment, and childcare to those addressing longer-term challenges such as climate change, public debt, and social welfare reform. Without the greater presence of younger politicians in elected office, the policies implemented by mostly older politicians may be detrimental to both the short- and long-term interests of younger generations.

In this article, I examine how the age of mayors in Japan affects the welfare policies they enact in office. Intergenerational conflict over welfare is an especially salient issue in many advanced democracies that confront the challenges of declining birth rates and aging populations, as shrinking workforces struggle to support rising welfare costs for growing elderly populations. Japan, as the world's oldest country, is at the forefront of this demographic

[1] Japan Municipal Elections Dataset (introduced later).

trend. It thus provides an ideal setting to analyze how politicians of different ages allocate scarce resources between competing priorities for social spending. On the one hand, spending on child welfare for young families is often viewed as a way to encourage fertility and support female labor force participation, thereby addressing the labor shortage and enhancing the long-term sustainability of the welfare system (Boling 2015; Iversen and Rosenbluth 2010). On the other hand, spending on elderly welfare can help governments keep up with the pressing financial and healthcare needs of an increasing number of senior citizens (Harper 2014; Muramatsu and Akiyama 2011).

I argue that younger politicians will be more likely than older politicians to promote spending on child welfare. In addition to testing this overall argument, I examine three potential mechanisms. First, younger politicians may be more motivated to advocate for child welfare spending because they are at a different life stage compared to older politicians—specifically, many are in the midst of raising young children themselves—which can significantly influence their policy priorities and preferences. Second, younger politicians may prioritize child welfare more due to their longer time horizons, recognizing that young people will disproportionately face the long-term consequences of declining birth rates on future economic growth and prosperity. Third, younger politicians may seek to enhance child welfare benefits because they are driven by stronger electoral incentives to respond to younger voters, who could form a critical support base for them in elections.

I test these hypotheses using a regression discontinuity design (RDD) and an original dataset of Japanese mayoral elections matched with detailed municipal expenditures data. Younger mayors are not randomly distributed across municipalities, which can differ in their welfare needs and preferences. By leveraging close elections between younger and older candidates, the advantage of my design is that I can estimate the causal effect of a municipality electing a younger mayor on social spending while holding constant the demographic composition, ideology, and socioeconomic characteristics of municipalities. Focusing on Japan offers further advantages because mayors have considerable discretion to implement their

preferred welfare policies, and the municipal expenditures data allow me to identify spending targeted at different age groups and time periods. However, there is a lack of data on municipal elections. To fill in this gap, I use web scraping to assemble an original dataset of 12,191 mayoral candidates competing in 6,215 elections (2006–2019). I then supplement the findings from this new dataset with insights from interviews with 15 mayors and 20 municipal bureaucrats working inside social welfare departments.

I find that municipalities that elect younger mayors increase their spending on child welfare, particularly through long-term investments in infrastructure, such as constructing daycare centers. Mechanism tests and interviews suggest the effects are driven more by intrinsic motivations—namely, the different life stage and longer time horizons of younger mayors—than by extrinsic electoral incentives to court younger voters. Importantly, while this increased emphasis on child welfare does divert resources from other policy areas, it does not result in cuts to elderly welfare. This underscores a nuanced understanding of intergenerational conflict, suggesting that greater youth representation can shape budgetary allocations benefiting young families without necessarily exacerbating tensions between the welfare needs of the young and old.

In sum, I provide evidence that young people's descriptive representation in elected office can lead to their substantive representation in social spending. In addition to adding age to the broader literature on elite characteristics and representation, I highlight two mechanisms unique to age: life stage and time horizons. In doing so, I show how younger politicians are more likely to allocate government resources both toward their younger constituents at a similar life stage and toward the future through greater long-term investment in child welfare infrastructure. My work also contributes to research on the age orientation of welfare programs, which has traditionally emphasized the role of electoral institutions (Estévez-Abe 2008; Lynch 2006). I find that even under the same majoritarian system, the age of politicians can make a substantial difference in spending that affects the public's well-being, especially individuals at the child-rearing life stage. My results further suggest that the extent to which

young people are represented in political institutions will have ramifications for how societies respond to the challenges posed by declining birth rates and aging populations.

The Consequences of Youth Underrepresentation

In most countries, elected officials tend to be older than most of the constituents they represent. Recently, this common feature of political institutions has attracted significant scholarly attention. However, research to date has predominantly focused on understanding the causes behind the descriptive underrepresentation of young people in elected offices. Scholars, for instance, point to factors such as lower political ambition among younger citizens (Lawless and Fox 2015; Shames 2017) and institutional barriers such as high minimum age requirements and electoral systems that favor more experienced candidates (Joshi 2013; McClean 2020; Stockemer and Sundström 2022). Conversely, there is less evidence that voter biases are to blame. In fact, most work finds voters harbor negative biases primarily against elderly candidates, but are broadly supportive of younger candidates (Eshima and Smith 2022; McClean and Ono 2024).

Despite this progress in identifying the causes driving young people's descriptive underrepresentation, there remains limited research on its potential consequences for the substantive representation of young people's interests in policy outcomes. On the one hand, there are reasons to believe that youth underrepresentation does have important implications for public policy. There is extensive evidence that politicians are more likely to advocate for policies important to those who share their personal characteristics, including their gender, race, class, and sexual orientation. Scholars reliably find that descriptive representatives provide greater substantive representation for groups with whom they share identities, whether measured by constituency service (Broockman 2013), speeches (Clayton, Josefsson and Wang 2017), bill sponsorships (Schwindt-Bayer 2006), roll-call voting (Carnes 2012), or even policy outcomes (Reynolds 2013), including social spending (Chattopadhyay and Duflo 2004). Although age remains understudied in this literature, we might anticipate that younger

politicians will similarly be more likely than older politicians to prioritize and advance the welfare interests of their younger constituents.

On the other hand, there are reasons to doubt whether the relationship between the descriptive and substantive representation of young people will be as robust as it is for other underrepresented groups. Within the broader literature on political representation, scholars frequently debate whether politicians advocate for their group's interests primarily due to *intrinsic motivations*—reflecting personal preferences, shared experiences, and feelings of group solidarity that foster sincere concern for the group's welfare—or due to *extrinsic motivations*, driven by electoral incentives and strategic calculations aimed at securing political support (Broockman 2013; Lowande, Ritchie and Lauterbach 2019; Mansbridge 1999).

Considering intrinsic motivations first, research on gender and race, for instance, highlights the distinct and persistent experiences associated with being a woman or a racial minority as a key mechanism behind substantive representation (Mansbridge 1999; Swers 2002). By contrast, youth is a universally shared and inherently temporary identity: all older representatives were once young themselves, and thus might plausibly claim to understand younger constituents' concerns based on their own past experiences. Additionally, the fleeting nature of youth may make the exclusion of young people from public office appear on its face to be less unjust compared to the systematic exclusion faced by women and minorities, groups that anticipate underrepresentation throughout their entire lives (Bidadanure 2021; Phillips 1995). Younger politicians, therefore, might feel comparatively less compelled by intrinsic motivations to advance youth-favored policies because they expect to eventually age out of this identity group.

Turning to extrinsic motivations, younger politicians' incentives to advocate for youth-oriented policies might also be tempered by the considerable electoral power of older voters. In many democracies, older people are not only becoming more numerous due to population aging, but also turn out to vote at higher rates, are better organized, and donate more to political campaigns than their younger counterparts (Goerres 2009; Holbein and Hillygus

2020; Vlandas 2018). The growing "grey power" of elderly constituents may discourage younger politicians from significantly reallocating social spending toward younger populations, as doing so could risk alienating this influential voting bloc. Consequently, younger politicians' support for youth-friendly policies may manifest primarily in less publicly visible forms of representation, such as constituency service (Broockman 2013), rather than overt policy decisions that involve highly visible reallocations of public resources.

Even if younger politicians possess intrinsic motivations, extrinsic motivations, or both, their relative lack of experience could limit their effectiveness in translating these motivations into substantive policy outcomes. Indeed, minimum age requirements for political offices are often justified by the belief that a certain level of life and professional experience is essential for effective governance (Bidadanure 2021; Stockemer and Sundström 2022). Although younger candidates have, by definition, reached the minimum age of eligibility, they may still lack the experience, institutional knowledge, and professional networks needed to successfully navigate complex political environments. Such experience can be especially consequential in legislative contexts, where seniority norms and established hierarchies frequently constrain the influence of younger representatives on policy decisions (Epstein et al. 1997). Similarly, in executive roles, younger politicians' relative inexperience may hinder their ability to effectively direct bureaucratic institutions and collaborate with older, more experienced politicians to realize their policy objectives (Lee and McClean 2022).

While there is limited research on the policy consequences of youth underrepresentation, the study closest to this one is Curry and Haydon (2018), who investigate how older lawmakers (aged 65 and over) influence bill introductions in the US Congress.[2] They find only small, conditional effects of age on substantive representation: older members of Congress are marginally more likely to introduce low-salience bills relevant to seniors, whereas for high-salience legislation, the determining factor is instead the size of the senior constituency in a lawmaker's district. However, these modest effects might reflect their choice to study older

[2]Alesina, Cassidy and Troiano (2019) do not focus on representation but find that younger elected officials are more likely to participate in political business cycles by increasing spending in election years.

people, a group already descriptively and substantively well-represented in politics. Their limited findings may also result from their exclusive focus on a single legislative setting, the US Congress, where representatives face significant cross-pressures from other lawmakers, party leadership, and the executive branch during the policymaking process.

This study empirically tests the unresolved question about whether young people's descriptive presence in political institutions can lead to their substantive representation in policy outcomes. Using original data and a novel approach, I estimate the causal effect of electing a younger representative on the allocation of social spending toward young families.

Younger Politicians and Social Spending

In many contexts, younger and older people have distinct preferences regarding how politicians should distribute limited government resources for social welfare.[3] Younger people typically favor greater public spending on childcare, parental leave, maternity care, and family allowances, which can ease their ability to raise children and remain in the labor force (Busemeyer, Goerres and Weschle 2009; Iversen and Rosenbluth 2010). Older people, in contrast, generally prefer increased spending on pensions, senior services, and healthcare, which disproportionately benefit elderly retirees (Goerres 2009; Vlandas 2018).

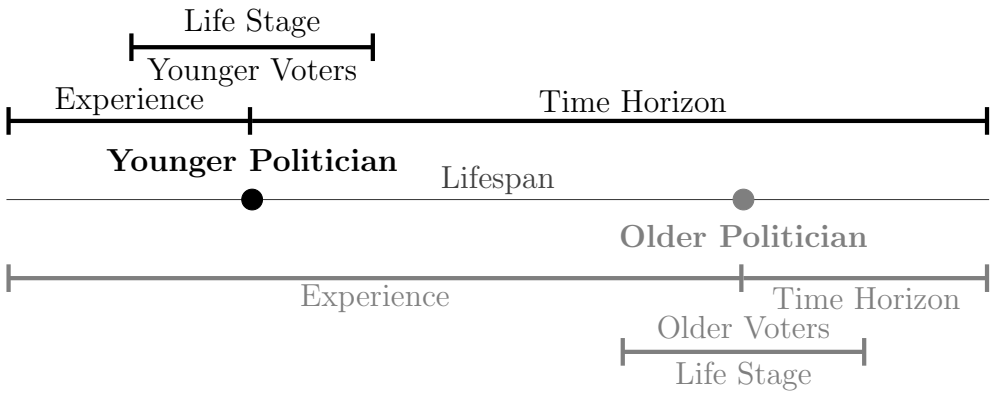
I argue that younger politicians will be more likely to represent the preferences of their younger constituents in their policy decisions. In other words, in line with findings on other social identities, my first expectation is that the descriptive representation of young people in public office will in fact translate into their substantive representation in social spending:

> **H1:** Younger politicians will be more likely than older politicians to increase spending on child welfare relative to spending on elderly welfare

In addition to evaluating the overall effect of younger politicians on social spending, I investigate three potential mechanisms underlying this relationship. To do so, I build on the

[3]The extent of intergenerational differences in welfare preferences varies by country and policy issue (Busemeyer, Goerres and Weschle 2009). We might expect young people's descriptive representation to matter more for substantive representation in cases where preferences differ more across age groups.

Figure 1: Younger and Older Politicians



intrinsic-extrinsic framework in the representation literature by highlighting the unique temporal dimensions associated with age as an identity characteristic. As illustrated in Figure 1, younger politicians differ from older politicians not only in their levels of experience but also in characteristics stemming from their current position along the lifespan. Although older politicians were once young, younger politicians are young *now*. This means they currently share a common life stage and longer time horizons—that is, they expect to have a longer remaining lifespan—with their younger constituents, creating distinct intrinsic motivations. Moreover, despite the greater electoral influence typically exercised by older voters, younger politicians' proximity in age to younger voters can still produce extrinsic motivations, by providing strategic electoral incentives to prioritize this group's interests.

First, younger politicians may be stronger advocates for child welfare spending due to intrinsic motivations based on their current life stage. Compared to older politicians, younger politicians are more likely to be raising young children, placing them in a similar situation to many of their younger constituents who are also raising families. This shared life stage may enable younger politicians to empathize more deeply with challenges such as restrictive parental leave policies, limited daycare availability, and rising childcare costs—issues directly relevant and personally meaningful to both themselves and the young families they represent.

Additionally, the nature of childcare challenges has evolved significantly over time. Young parents today typically express much greater demand for public childcare support than earlier

generations, driven largely by the rise of dual-earner households and shifting societal expectations around work and family balance (Boling 2015; Fukai 2017; Iversen and Rosenbluth 2010).[4] As a result, younger politicians raising children may feel a stronger personal connection with fellow young parents, whereas older politicians' past experiences might provide fewer insights into contemporary childcare policy challenges.

If shared life stage serves as an important mechanism, then we might expect to observe stronger effects among younger politicians actively raising children compared to their counterparts who are not:

> **H2:** Younger politicians raising children will be more likely than younger politicians without children to increase spending on child welfare

Second, younger politicians may also have stronger intrinsic motivations to advocate for child welfare spending because of the longer time horizons they share with their younger constituents.[5] Policies aimed at child welfare, such as investments in childcare and early childhood education, are often thought to yield significant long-term societal benefits by enhancing human capital, productivity, and overall economic prosperity in the decades ahead (Esping-Andersen 2009; Heckman 2006). Younger politicians, recognizing that these benefits will disproportionately accrue to members of their age group and their children, may thus feel more personally driven to invest in child welfare—particularly compared to older politicians who are less likely to directly experience these longer-term social gains.[6]

These motivations may be especially pronounced in advanced democracies such as Japan that are confronting the severe long-term challenges posed by declining birth rates and population aging. These demographic trends threaten future economic growth, fiscal stability,

[4] Although this study focuses specifically on contemporary age effects, future research could explore generational differences in policy advocacy across different time periods and age cohorts.

[5] While research linking politicians' ages to their time horizons is limited, studies suggest that younger people often exhibit lower risk aversion and a greater openness to policies with longer-term payoffs than older people (Alesina and Passarelli 2019; Albert and Duffy 2012).

[6] Although research shows that grandparents can also exhibit altruistic behaviors toward their grandchildren's generation (Silverstein and Giarrusso 2010), I expect that older politicians' shorter time horizons may nonetheless weaken their motivation to prioritize policies with longer-term payoffs.

and the sustainability of the welfare system—burdens that stand to be disproportionately borne by younger generations in the form of higher taxes, reduced social benefits, and limited economic opportunities (Harper 2014; Muramatsu and Akiyama 2011; Lynch 2006). Recognizing this looming shared future, younger politicians may be especially motivated to invest more in child welfare today, aiming to mitigate these long-term burdens through policies designed to encourage fertility, increase labor force participation, and ensure the resilience of social welfare systems for their younger constituents in the coming decades (Boling 2015; Fukai 2017; Iversen and Rosenbluth 2010).[7]

If longer time horizons represent a key mechanism, we should observe younger politicians not only spending more on child welfare overall but also demonstrating a greater willingness to enact policies with potentially longer-term payoffs. Politicians regularly face what Jacobs (2011) calls an "intertemporal dilemma," as they are torn between two competing goals: on the one hand, minimizing taxes and maximizing immediate benefits through *short-term spending*; and on the other, allocating resources toward *long-term investments*, such as expanding childcare infrastructure, which entail upfront costs but offer the prospect of greater social returns in the long run.[8] Recognizing their younger constituents' greater stake in future challenges such as population aging, younger politicians may therefore be more willing to accept the intertemporal trade-off that comes with longer-term investments.[9]

> **H3:** Younger politicians will be more likely than older politicians to increase long-term investment in child welfare relative to short-term spending

[7]Although evidence regarding the effectiveness of such policies—especially for increasing fertility—is mixed, interviews (discussed later) suggest that younger politicians often share a belief with their younger constituents that these measures are important and necessary responses to demographic decline.

[8]Previous research often portrays reelection-minded politicians as gravitating toward short-termism unless institutions such as term lengths, seniority systems, or safe seats extend their expected time horizons in office (Dal Bó and Rossi 2011; Jacobs 2011; Simmons 2016). Some scholars thus suggest that older politicians might be more favorable toward long-term policies because they are more likely to represent safer districts and benefit from seniority systems (Kato 1994; Naoi 2015). Holding institutional conditions constant, my expectation is that younger politicians will be more likely to increase long-term investment in child welfare.

[9]Recent work in the U.S. also finds evidence of an emerging age-based identity among young people, especially regarding long-term challenges such as climate change (Munger 2022; Trachtman, Anzia and Hill 2023). My argument suggests younger politicians might similarly be more willing to address climate change due to the shared longer-term concerns they have with their younger constituents.

Finally, a third possible mechanism is that younger politicians could be more inclined to increase child welfare spending due to extrinsic motivations based on strategic electoral calculations. Younger politicians' proximity in age to younger voters may lead them to perceive this group as a particularly valuable electoral base, especially given the relative challenge of securing support from older voters, who often have established loyalties to more senior politicians. Consequently, younger politicians might strategically adopt child welfare policies specifically tailored to appeal to younger voters, believing that mobilizing support among their generational peers may represent their most viable electoral strategy.

Several factors may enhance the strategic attractiveness of younger voters for younger politicians. Although younger voters generally turn out at lower rates and donate less to political campaigns (Goerres 2009; Holbein and Hillygus 2020), their relative lack of attachment to specific politicians and broader disaffection with traditional politics could make them particularly receptive to targeted mobilization or persuasion (Franklin 2004). Younger politicians, in particular, may have a comparative advantage in capitalizing on this strategic opportunity, as their age allows them to present themselves as more authentic and credible representatives of younger voters' interests and as potential challengers to political establishments typically dominated by older politicians. Indeed, research across a variety of settings finds that younger voters often show clear preferences for younger candidates at the ballot box (Sevi 2021; McClean and Ono 2024; Pomante and Schraufnagel 2015), further reinforcing younger politicians' incentives to mobilize this constituency.

Alternatively, electoral incentives for younger politicians to expand child welfare could also arise from voter-initiated dynamics. While there is limited direct research on age specifically, studies indicate voters often contact representatives who share their descriptive characteristics (Broockman 2014; Gay 2002), suggesting younger voters might similarly engage more with younger politicians. The emergence of a younger politician might therefore galvanize previously disengaged young voters to more actively demand increased child welfare spending, creating external incentives for younger politicians to respond accordingly.

If younger politicians indeed face stronger electoral incentives to appeal to younger voters, we might expect their support for increased child welfare spending to depend on the relative share of younger voters in their constituencies:

> **H4:** Younger politicians in younger constituencies will be more likely than younger politicians in older constituencies to increase spending on child welfare

Research Design

Japan offers an ideal setting to test these hypotheses because it combines a highly salient policy context—defined by severe demographic pressures and the potential for intergenerational conflict over welfare—with an institutional structure in which directly elected mayors have both the autonomy and authority to shape local social spending. Focusing on mayors thus allows me to examine how a politician's age may influence social spending in a setting where both the stakes and the executive's discretion over welfare policy are especially clear.

Due to its declining birth rate and rapidly aging population, social welfare has become one of the most pressing issues in Japanese politics. Since 1970, the share of the population aged 65 and over has quadrupled, rising from 7.1% to 28.4%. Over the same period, the share of children under 15 has been cut nearly in half, falling from 24.0% to 12.1% (Ministry of Internal Affairs and Communications 2020*a*).[10] Assuming these trends continue, overall social welfare expenditures—already the largest part of the government's budget—are projected to increase by 60% in absolute terms over the next two decades, while the number of working-age people is expected to decline by 20% (Cabinet Office of Japan 2018).[11] Understanding how politicians allocate welfare resources between people at different life stages is thus crucially important, not only because it affects individual decisions about family planning, savings, investment, and retirement, but also because it has broader consequences for Japan's labor market, economic growth, and fiscal sustainability (Iversen and Rosenbluth 2010; Lynch 2006; Rosenbluth 2007).

[10] While there were nine working-age people per senior citizen in 1970, today there are just two.
[11] Japan's total population is also expected to decline from 127 million to 107 million by 2040.

Although the demographic challenges that Japan faces loom large, many scholars agree that the country has been slow to reform the welfare system, that the current system favors the elderly, and that, as a result, many young people are concerned with intergenerational inequality (Estévez-Abe 2008; Kweon and Choi 2022; Lynch 2006; McClean and Ono 2024; Umeda 2022). Some studies have suggested that Japan's elderly-biased welfare system may be related to the large elderly share of the population (Eschker 2003), but others have failed to find a clear link between population aging and expenditures on elderly welfare (Lynch 2006; Tepe and Vanhuysse 2009). In contrast, other scholars have instead pointed to political institutions such as welfare regimes and electoral systems as the primary reasons for why countries like Japan have devoted more benefits toward the elderly than young families (Boling 2015; Estévez-Abe 2008; Lynch 2006).

While prior work has focused on Japan's national government, mayors are centrally involved in nearly every aspect of welfare policy, with the exception of pensions, and municipalities distribute more than half (57%) of all welfare expenditures (Ministry of Internal Affairs and Communications 2020*b*).[12] Higher tiers of government do require that municipalities provide a minimum level of welfare services, but mayors otherwise have significant powers to implement their preferred welfare agenda. In principle, mayors share budget-making responsibilities with municipal assemblies, but in practice, they have broad authority over social spending (Bessho 2012; Tsuji 2017).[13] Mayors are also relatively free to enact their preferred policies because parties are less involved in local than in national politics, and nearly all mayors run as independents during elections. Additionally, over 99% of mayors are ethnically Japanese and 98% are men, which allows age to emerge as one of the most meaningful sources of identity-based variation among elected officials. Taken together, these features make it easier to estimate the effect of a politician's age on social spending without

[12]Although private companies also provide welfare services, most age-related care is either provided or heavily subsidized by the government (Estévez-Abe 2008).

[13]Mayors have the exclusive right to draft the annual budget and can introduce legislation, veto assembly resolutions, and modify the budget when the assembly is not in session.

having to account for significant constraints imposed by other political actors.[14]

Studying the link between the age of mayors and social spending also offers an opportunity to address a puzzle not readily explained by existing theories. Under Japan's unitary government structure, mayors across the country have similar institutional powers and are elected through the same plurality system to serve four-year terms with no term limits. Prior research on institutional incentives would thus suggest that these mayors should pursue similar welfare policies (Estévez-Abe 2008; Lynch 2006). In contrast to these expectations, however, Japanese mayors differ significantly in the level of welfare benefits they provide for young families and the elderly, even across municipalities that otherwise share very similar income levels and age demographics (Fukai 2017).

In fact, this combination of mayoral budget authority and variation in municipal welfare services often puts mayors on the front lines of responding to perceived generational inequalities in social welfare. In particular, many young families complain about the substantial variation in childcare availability across municipalities. Despite Japan's declining birth rate, demand for publicly provided childcare has risen sharply in recent years due in large part to the increased participation of women in the workforce (Fukai 2017; Rosenbluth 2007). While some mayors have managed to keep up with demand through sustained investment in childcare infrastructure, others have made only modest increases in overall spending, or have prioritized more immediate forms of support, such as subsidies to reduce the cost of access. As a result, long waitlists for public daycare have become a major issue of contention in many municipalities, with some estimates suggesting that as many as 600,000 to 850,000 children have been unable to find spots in recent years (Funakoshi 2013).[15]

The idea that younger mayors in particular promote investing in children matches well with the observed behavior of several mayors who have recently attracted national attention. For instance, when Shuhei Azuma, 28, managed to defeat a 61-year-old incumbent in Shijon-

[14]In contrast, measuring the effect of a legislator's age on their social policies is more challenging because they have to work closely with other legislators, their party, and often the executive to pass legislation.

[15]The Japanese government typically estimates that only 30,000 to 45,000 children are on daycare waitlists, although this number does not include parents who have given up waiting.

awate to become the youngest mayor in Japan, he differentiated himself from his opponent during the campaign primarily by pledging to make increased support for child-rearing the center of his policy platform. Naomichi Suzuki, 31, in his first year as mayor of Yubari, abolished the vice mayor position and pressured the central government to restructure the aging mining town's debt repayment schedule to free up money to fund medical care for the city's infants. Suzuki went on to become Japan's youngest governor in 2019. Finally, Naomi Koshi, 37, as mayor of Otsu, built 20 new nurseries to house 2,000 children, a program that successfully eliminated long waitlists for daycare spots in her city. She is also the first mayor in Japan to institute mandatory parental leave for both male and female city employees.

Japan Municipal Elections Dataset

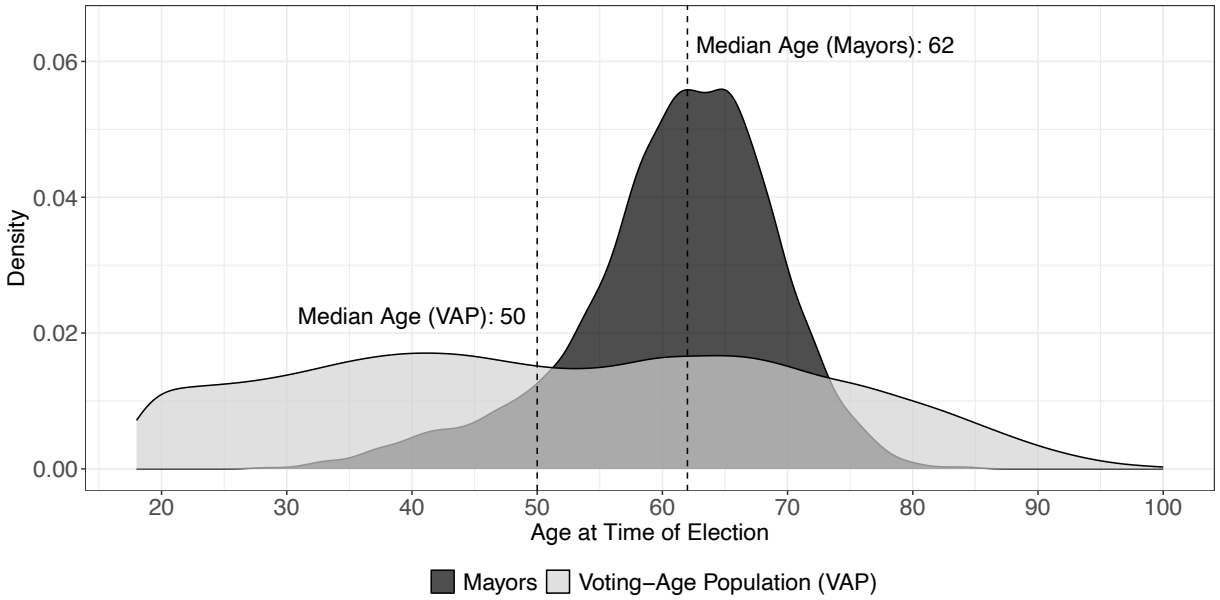
One challenge to the study of local politics in Japan has been the lack of existing datasets on either candidates or outcomes in municipal elections. This absence of systematic data on municipal politicians makes it impossible for researchers to answer questions such as whether the age of mayors affects social spending.

I fill in this gap by building an original, candidate-level dataset of municipal elections held between 1999 and 2021, which I call the Japanese Municipal Elections Dataset (JMED). To construct JMED, I first used web scraping to collect information on elections from a variety of online platforms in Japan that seek to aggregate data from newspapers, election returns, individual users, and candidates themselves. I then supplemented this information by searching through newspaper archives as well as candidate and municipal websites to correct errors and further fill out the demographic information of candidates.[16]

In this article, I focus only on mayoral elections held from 2006 to 2019, which include 6,215 elections across Japan's 1,741 municipalities. I exclude elections between 1999 and 2005, a period in which a significant number of municipalities merged, and also those held after 2019, because of the COVID-19 pandemic's impact on elections and social spending.

[16]See Appendix Section 1 for additional information on the construction of JMED.

Figure 2: Age Distribution of Voters and Mayors in Japan



Notes: The figure shows the age distribution of the voting age-population (grey) and elected mayors (black) in Japan (2006–2019). An individual must be 18 to vote and 25 to run for mayor. Sources: JMED; Ministry of Internal Affairs and Communications (2020*a*).

Both of these factors complicate efforts to link the age of mayors to their social policies.[17]

My final dataset includes information on the names, ages, gender, incumbency, partisanship, and vote totals for the 12,191 candidates who ran for mayor between 2006 and 2019. As discussed later, I further compile biographical information for a subset of politicians who competed in close races between younger and older candidates, including their education, family structure, party support, and prior experience in politics or government.

Figure 2 compares the age distribution of mayors to that of the voting-age population. To run for mayor, an individual must be at least 25 and eligible to vote in the municipality. Mayors elected during this period range in age from 28 to 85, although most are much older than the minimum age requirement. The median age among mayors is 62 at the time of their election, 12 years older than that of the voting-age population.[18]

Mayors thus tend to be older than most of their constituents. As Figure 2 shows, the

[17]The Junichiro Koizumi administration also implemented the "Trinity Reforms" in the early 2000s, which changed the nature of municipal finances, including how they funded social welfare programs.

[18]The median age of candidates is also 62, and the age distribution is very similar to Figure 2 (Figure A1).

greatest gap in descriptive representation occurs for people under 50, who make up half the voting-age population but just 10% of mayors. By contrast, people in their late 50s, 60s, and early 70s are significantly overrepresented in mayor's offices. For example, the number of mayors elected between the ages of 60 and 65 (2,030) during this period was almost 700 times greater than the number of mayors under 30 (three), over 18 times greater than the number under 40 (112), and nearly four times greater than the number under 50 (557).[19]

While young people are underrepresented in mayoral positions, many in Japan have experienced living in a municipality governed by a younger mayor: 344 (20%) of Japan's 1,741 municipalities (spanning across 46 of 47 prefectures) elected at least one mayor under 50 between 2006 and 2019.[20] Compared to municipalities with mayors in their 50s or older, municipalities with younger mayors tend to have younger populations, are more populous, are more likely to be located in metropolitan areas, and have spent less per capita on investing in both child and elderly welfare.[21] The decisions made by mayors in their 20s, 30s, and 40s have thus affected a large swath of Japan's population, including approximately 48 million (38%) of the country's 126 million residents.

Municipal Social Welfare

To study how a mayor's age affects their social spending, I merge JMED with detailed municipal budget reports published by the Japanese government (Ministry of Internal Affairs and Communications 2020*b*).[22] A benefit of this data is that municipalities distinguish between discretionary spending, which is allocated during the annual budget process, and mandatory spending, which is typically ongoing and includes entitlement programs set by the central government (such as the child allowance system), interest payments on debt,

[19]The median age of mayors increased from 60 in 2006 to 64 in 2018 (Figure A2). While the share of mayors under 50 rose slightly, the share aged 60 and older increased more substantially (Figure A3).

[20]More than twice as many municipalities (739, or 42%) had at least one candidate under 50.

[21]Mayors under 50 are more common in the Kanto and Kansai regions, which include Japan's two largest metropolitan areas surrounding Tokyo and Osaka, respectively, but are less common in the less dense regions of Hokkaido, Tohoku, and Shikoku (Figure A5, Table A1).

[22]See Appendix Section 2 for more information on the municipal budget data.

and salaries for government personnel.[23] This level of detail enables me to focus only on discretionary spending, where the mayor has more influence.[24]

Another benefit is that municipalities use discrete budget categories to account for discretionary social spending targeted at different age groups. The "Child Welfare" category includes spending on parental leave benefits, centers for maternity and daycare support, subsidies for parents, and benefits for children with disabilities. The national government disburses pensions, but the "Elderly Welfare" category includes public expenditures on nursing services, subsidies, and facilities for the elderly.[25]

To test the time horizons mechanism (H3), I use the two largest discretionary spending categories—"Subsidies" and "Investment"—as proxies for short-term spending and long-term investment.[26] Subsidies typically take the form of direct transfers to individuals or organizations, including funds that reduce the cost of accessing welfare services or support programs not mandated by central government formulas. These allow mayors to deliver quick social gains to their constituents. In contrast, the investment category consists almost entirely of public works projects to construct, expand, or upgrade childcare and elderly care facilities. These projects are costly in the short term and often span multiple mayoral terms, but offer the potential for greater long-term social returns by expanding the quality and capacity of local welfare infrastructure.

Given the wide variation in social spending and population demographics across municipalities, I focus on the natural logarithm of spending per capita to reduce the possibility that a few outliers could drive my results.[27] Using the government's age cutoffs for who qualifies for welfare benefits, I divide child welfare expenditures by the population under 15 years old and elderly welfare expenditures by the population 65 years and older.

[23] Under the central government's child allowance system, parents of children under 15 can receive monthly payments (*jidou teate*). While the rules governing these payments have changed over time, the amount that parents receive depends on the number of children, their age(s), and the household's income level.

[24] Municipal spending on welfare is typically evenly divided between mandatory and discretionary spending.

[25] Although Japan's overall welfare system may be biased toward the elderly, municipalities on average spend more per capita on child welfare than elderly welfare, though there is significant heterogeneity.

[26] See Appendix Section 2.4 for more information on these categories and a discussion of alternative proxies.

[27] The results are similar for non-logged values of spending.

Regression Discontinuity Design

Identifying the causal effect of a municipality electing a younger mayor on social spending is no easy task. As discussed earlier, municipalities with younger mayors differ in observable characteristics (e.g., population size and age demographics) from those with older mayors. While these factors could be controlled for using multivariate regression, it is unlikely that such an analysis would be able to account for other, unobservable differences between these municipalities that could affect both the age of mayors and welfare expenditures.

To address these concerns, I use a regression discontinuity design (RDD) to analyze close elections between younger and older mayoral candidates. The core assumption of this design is that in close single-member district elections, in which the winner changes discontinuously at 50% of the top-two candidate vote share, which candidate wins is thought to be as-if randomly assigned so long as there is some unpredictability in the ultimate vote (Lee 2008). Because of this as-if random assignment, municipalities that narrowly elect the younger candidate should be largely similar in both observable and unobservable characteristics to those where the older candidate barely wins, except for the age of the elected mayor. As a result, this design enables me to causally identify the local average treatment effect (LATE) of a municipality narrowly electing a younger mayor, compared to narrowly electing an older mayor, on future outcomes such as social spending.[28]

While close-election RDDs have been used to estimate the effects of municipalities electing mayors that vary by gender (Ferreira and Gyourko 2014), partisanship (De Benedictis-Kessner and Warshaw 2016), and race (Hopkins and McCabe 2012) on policy outcomes, one challenge to adapting this design to study age is that identifying who qualifies as a relatively "young" candidate can vary depending on the cultural and institutional setting (Joshi 2013; Stockemer and Sundström 2022). In the context of Japanese mayoral elections, I begin with an age cutoff of 50 because it has several nice features. As discussed earlier, 50 represents

[28] An important distinction is that this design enables me to identify the effect of electing a younger mayor, holding constant other characteristics of the municipality, but not the effect of a mayor's age, holding constant other characteristics of the mayor (Bertoli and Hazlett 2025; Marshall 2022).

the median age of the voting-age population and the youngest 10% of elected mayors (Figure 2). A cutoff at 50 also marks a salient age for preferences regarding social spending, as candidates in their late 20s, 30s, and 40s represent the typical age range of young parents who receive child welfare benefits.[29]

In the main analyses, I estimate the effect of narrowly electing a mayoral candidate under 50 rather than a candidate 50 or older on social spending within the sample of municipalities with close elections between these two candidates, where the median age gap between candidates is 20 years.[30] After presenting these results, I then test how the effect size varies across alternative age cutoffs for younger mayors. Following the best practices outlined in Cattaneo, Idrobo and Titiunik (2019), I model these relationships using local linear regression, an optimal bandwidth chosen to minimize the mean square error, standard errors clustered by municipality, and robust confidence intervals adjusted to account for remaining bias. To increase statistical efficiency, I estimate the effect of a mayor's age on changes in spending rather than on levels (Lee and Lemieux 2010). The results for the first difference of my logged dependent variable can be interpreted approximately as the percentage change in expenditures due to electing a mayor under 50. My main analyses focus on the differences in spending on child and elderly welfare between the year before the election and the year after the election.[31]

Interviews With Mayors and Bureaucrats

Finally, I draw on interviews conducted with mayors and municipal bureaucrats during fieldwork in Japan between October 2018 and July 2019 to supplement the RD analyses. Interview subjects were randomly selected via JMED, although I oversampled municipalities

[29]See Appendix Section 3.2 for more details on modeling age as a binary treatment.

[30]The LATE is thus only estimated for municipalities with close elections where the top-two candidates feature one candidate under 50 and one candidate 50 or older. This excludes elections where a younger or older candidate comfortably wins, elections where a single candidate wins unopposed, and elections where the top-two candidates are both younger or both older. See Appendix Section 3.1 for more details.

[31]Following De Benedictis-Kessner and Warshaw (2016), I focus on the year after the election to balance between giving the new mayor time to have an impact on the budget and endogenous responses from other political actors to the mayor's influence that may arise later in the term.

that met three criteria: i) municipalities with younger mayors (under 50 at the time of their election); ii) municipalities with mayors whose election was part of my RD analyses; and iii) municipalities within a few hours train ride from Tokyo or Osaka (for practical purposes). I contacted the offices of local officials by email to request an appointment and continued until I reached my target goal of at least 30 participants. In total, I interviewed 15 mayors and 20 bureaucrats working in social welfare departments across 20 municipalities. All interviews were conducted in Japanese and lasted an average of 45 minutes.[32]

Results

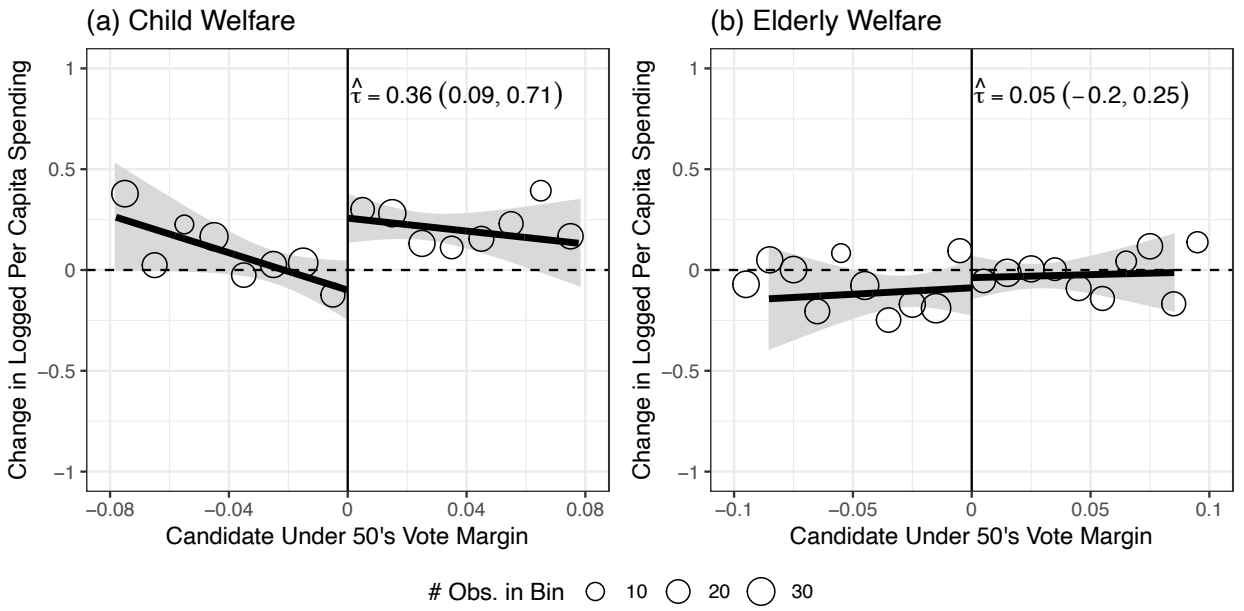
Do municipalities that elect younger mayors allocate more government resources than those with older mayors toward meeting the welfare needs of their younger constituents? Figure 3 presents a graphical representation of the main RD results.[33] The results for child welfare are in panel (a) and for elderly welfare in panel (b). The y-axis in each plot is the change in logged per capita spending for each welfare category from the year before to the year after the mayoral election, and the x-axis is the vote margin for the candidate under 50 in that election. The circles represent bins of the raw data and are sized according to the number of observations. On each side of the hypothesized discontinuity, where the margin of victory is equal to zero, I fit lines using a local linear smoother. The grey shaded areas represent 95% confidence intervals. Lastly, the RD estimate ($\hat{\tau}$) is provided for each figure with robust confidence intervals in parentheses.

Looking first at panel (a), I find that municipalities that elect younger mayors increase discretionary spending on child welfare relative to those with older mayors. Visually, this effect of electing a mayor under 50 can be seen in the large positive "jump" between the two fitted lines at the election threshold. Municipalities that narrowly elect the candidate under 50 increase their spending on child welfare by about 36% relative to municipalities that barely elect the candidate 50 or older. This increase represents a little over one half of a

[32]See Appendix Section 4 for more details on interviews.

[33]I take inspiration from De Benedictis-Kessner and Warshaw (2016) in plotting my RD results.

Figure 3: Effect of Electing a Mayor Under 50 on Social Spending



Notes: The figure presents the RD estimates ($\hat{\tau}$) of electing a mayor under 50 on the change in logged per capita discretionary spending on (a) child welfare and (b) elderly welfare.

standard deviation, an effect size typically viewed as moderate (Cohen 1992). In the average municipality in the RD analysis, it is equivalent to spending nearly 21,000 yen ($192) more per child under 15 or 365 million yen ($3.4 million) more in total.[34]

By comparison, panel (b) indicates that the election of a mayor under 50 does not lead to a corresponding decrease in a municipality's discretionary spending on elderly welfare. Unlike panel (a), there is no noticeable jump at the election threshold. The RD estimate is actually positive, although it is not statistically significant.

Do these effects vary greatly depending on the selected age cutoff for a younger mayor? In the Appendix, I find that the results are largely consistent across alternative modeling choices (Figure A6). The RD estimates of younger mayors on child welfare spending remain statistically significant for cutoffs between the ages of 41 and 55, with the effect size peaking at 48% (age 43), although below age 41 the number of observations starts to become too small to reliably estimate an RD effect. Similarly, the estimated effect of electing a young

[34]Conversions from yen to dollars are calculated using the average exchange rate in 2019 (1$ = 109¥).

mayor on elderly welfare spending remains insignificant across age cutoffs. Additional tests also show that the results are not driven by candidates close to one another in age who happen to fall just on either side of the chosen age cutoff (Table A5).[35]

Together, these results offer support for H1. Municipalities with younger mayors are more likely than those with older mayors to shift social spending toward greater benefits for young families. However, it is notable that this shift is primarily driven by younger mayors increasing spending on child welfare rather than taking support away from elderly welfare.

Mechanisms

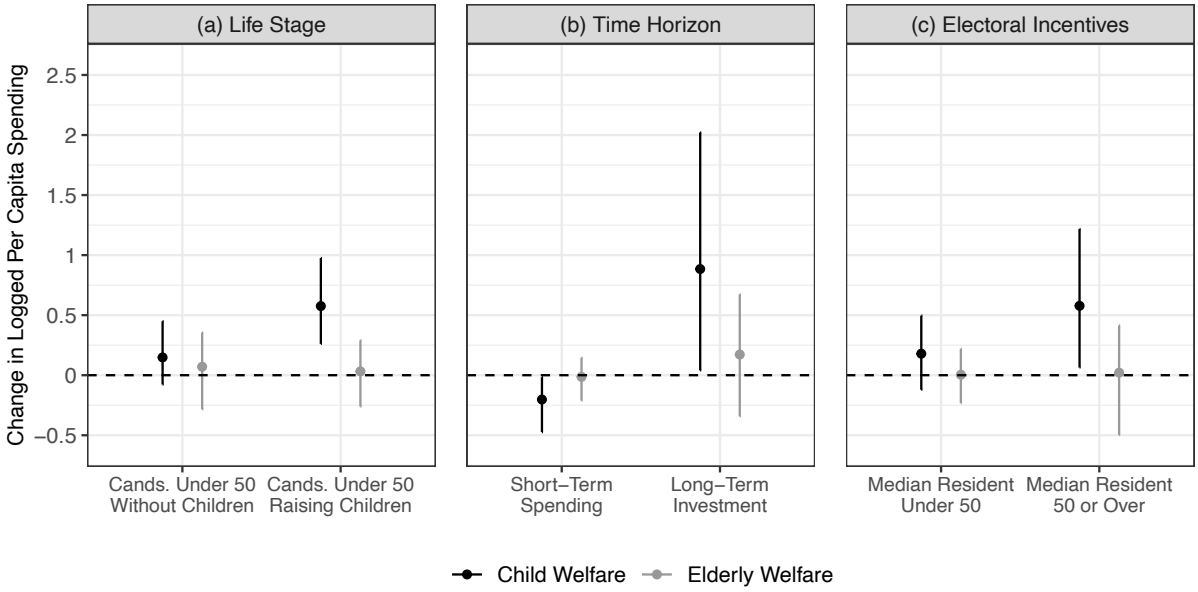
Next, I turn to evaluating the three proposed mechanisms. For life stage (H2), I split the main RD sample into two subgroups based on whether the younger candidate has children. Thus, one subgroup compares spending decisions by narrowly elected younger candidates with children to those by narrowly elected older candidates, while the other compares spending decisions by narrowly elected younger candidates without children to narrowly elected older candidates.[36] For time horizons (H3), I continue with the full set of elections and contrast the effect of narrowly electing a mayor under 50 on short-term spending with the corresponding effect on long-term investment. Finally, for electoral incentives to target young voters (H4), I split the main RD sample into two subgroups based on whether the median age of residents in the municipality is under 50. Thus, one subgroup compares narrowly elected younger and older candidates in municipalities with relatively more younger voters, while the other compares them in municipalities with relatively more older voters.

Figure 4 shows the results for child welfare (black) and elderly welfare (grey) spending. To conserve space and ease interpretation, I plot only the RD estimates and robust 95% confidence intervals for each analysis across three separate panels. As with the earlier analyses, these models all use local linear regression, an optimal bandwidth selected to minimize the mean square error, and standard errors clustered by municipality.

[35] See Appendix Section 3.5 for additional robustness checks.

[36] The older candidates in these elections may or may not have children; the subgroup definition is based solely on the younger candidates' parental status.

Figure 4: Mechanisms



Notes: The figure presents the RD estimates of electing a mayor under 50 on the change in logged per capita spending on child welfare (black) and elderly welfare (grey) across three panels. Panel (a) divides the main sample by whether the candidate under 50 has children. Panel (b) compares expenditures that were directed toward short-term spending with that toward long-term investment. Panel (c) divides the main sample by whether the median age of residents in the municipality is under 50.

Beginning first with child welfare expenditures, panel (a) indicates that whether a younger mayor is raising young children has consequences for their spending priorities (H2). Mayors under 50 raising children increase spending on child welfare by approximately 58%. This estimate is over three times larger than that for mayors under 50 without children (15%), and the difference between the two estimates is statistically significant.

Second, panel (b) suggests that the longer time horizons of younger mayors are also important (H3). Municipalities that elect mayors in their 20s, 30s, and 40s increase their long-term investment in child welfare by 88% relative to those with mayors 50 and older. In doing so, municipalities with younger mayors also appear to decrease short-term spending by about 20% relative to those with older mayors.[37] These results suggest that younger mayors may even be willing to reduce short-term benefits for young families in order to better fund long-term investments in infrastructure.

[37]These correspond to medium-size standardized effects of 0.51 and 0.34 standard deviations, respectively.

Third, panel (c) does not find that having a relatively greater number of younger voters in the municipality increases the effect of younger mayors on child welfare spending (H4). Contrary to the hypothesized relationship, the RD estimate is actually larger for mayors under 50 when the median age of residents in their municipality is 50 or over (58%) than when the median age of residents is under 50 (18%). While the estimate for municipalities where the median resident age is 50 or over is statistically significant, the difference between the two groups is not.[38]

Finally, it is notable that the effect of younger mayors on elderly welfare spending remains insignificant across the subgroups featured in panels (a), (b), and (c). In other words, whether a younger mayor is raising young children, is focused on short-term spending or long-term investment, or is in a younger or older municipality does not have a differential effect on the attention they pay to elderly welfare compared to older mayors.

In sum, I find more evidence for intrinsic motivations than extrinsic ones. The results suggest that younger mayors increase child welfare spending primarily due to personal concerns for the welfare of others in their age group—such as being in a similar life stage or sharing longer time horizons with their younger constituents—rather than because of stronger electoral incentives to appeal to young voters.[39]

Evidence from Interviews with Mayors and Bureaucrats

Lastly, interview evidence collected during fieldwork in Japan offers further insights into the findings from the quantitative analyses.

The idea that a mayor's age can shape their social policies resonated with both the mayors and municipal bureaucrats who spoke with me. In fact, local officials were near unanimous in emphasizing the central role of mayors in determining a municipality's welfare expenditures. As one mayor put it, "Mayors have four key powers concerning welfare policy: making policy decisions, directing human resources to carry out that decision, organizing the budget

[38]The interaction effect remains insignificant if the continuous version of resident median age is used instead.

[39]See Appendix Section 3.6 for additional tests aimed at ruling out alternative explanations.

to fund it, and, perhaps most critically, convincing the public that the policy is necessary."[40] In another interview, local bureaucrats praised the leadership of their younger mayor in successfully increasing investment in child welfare, "from finding the necessary funds, land, and teachers to open a new childcare center to addressing concerns from nearby residents about increased noise and traffic."[41] Other bureaucrats spoke about daycare projects that "never got past the planning stages or failed partway through" in neighboring municipalities, where older mayors were "not sufficiently concerned with the daycare shortage problem to take on the political risk of upsetting older homeowners."[42]

As for potential mechanisms, the importance of a mayor's life stage came through often in my interviews. As one younger mayor succinctly put it, "I feel that my age gives me insights into the challenges faced by young parents. I myself had difficulty finding a daycare spot for my child, as did many of my friends."[43] In contrast, the older mayors I met with seemed to draw different conclusions from their own experiences as parents. One older mayor, for instance, acknowledged that they did not understand "why young people are complaining about a lack of child care support" given that the "current welfare benefits are already much more generous than they were when we raised our children."[44]

In many of my interviews, when mayors discussed how their ages affect their preferences and priorities, their focus was more on time horizons rather than the allocation of welfare between age groups. Younger mayors reported feeling that their age gives them a "stronger sense of crisis about Japan's population shrinking in the future."[45] They talked about "implementing policies that are sustainable for the next 20 to 30 years" and the "downstream benefits to investing in child welfare not only for the low birth rate problem, but also for increasing the number of women who work and encouraging more people to move to [their]

[40] Author interview, October 25, 2018.
[41] Author interview, October 18, 2018.
[42] Author interview, July 10, 2019.
[43] Author interview, July 8, 2019.
[44] Author interview, July 11, 2019.
[45] Author interview, October 31, 2018.

municipality."[46] Older mayors, on the other hand, felt that their "age and life experience gave [them] a clearer sense of what is achievable within a mayor's term," and said that they could have the "greatest impact by focusing on the immediate needs of their constituents" in confronting the challenges associated with an aging population.[47]

Finally, interviews also helped to shed light on some of the findings that contradicted my initial hypotheses. For example, regarding the null effect of younger mayors on elderly welfare spending, every mayor spoke openly about the disproportionate power of older voters in elections and the difficulty in attempting to transfer welfare resources from the elderly to young families. One younger mayor even described directly asking a group of older residents if they would be willing to accept less welfare to free up funds for children but found "such strong opposition that [they] decided to find money in other areas of the budget instead."[48] As for the lack of support for the electoral incentives mechanism, several younger mayors that I interviewed did recognize that they have a comparative advantage in appealing to younger voters in elections, such as through their greater use of social media.[49] At the same time, even younger mayors acknowledged that they were contacted much more often by older voters, that it was mostly older voters who turned out at their campaign events, and that it was mostly older constituents who were members of their official support groups (*koenkai*).[50]

Discussion

There is significant evidence that a politician's race, gender, class, and sexual orientation can affect the extent to which they advocate on behalf of groups who share similar characteristics. In this article, I provide evidence that the descriptive representation of young people in political institutions can also matter for the substantive representation of their interests in policy outcomes. The election of a younger mayor in Japan can lead municipalities to

[46] Author interviews, October 11 and 15, 2018.
[47] Author interviews, October 5 and 26, 2018.
[48] Author interview, October 24, 2018.
[49] Author interview, July 2, 2019.
[50] Author interviews, October 24, 2018; July 8, 2019.

greatly increase their discretionary spending on child welfare, especially through long-term investments in young families, but does not appear to affect elderly welfare spending.

These findings suggest that the age bias of political institutions deserves further attention, particularly in societies that confront the challenges of declining birth rates and aging populations. On the one hand, individuals who are worried about intergenerational conflicts over social welfare might view the asymmetric effects of a politician's age on welfare policies as good news. Younger mayors do not seek to defund programs for the elderly to increase benefits for young families. On the other hand, the results and interviews reinforce past studies on the disproportionate power of the elderly in elections, casting doubt on whether young people can garner sufficient support for issues that are important to them in settings where they lack descriptive representation—as is often the case.

This article also contributes to the literatures on elite behavior and representation by showing how a politician's characteristics can affect how they allocate government resources not only between groups with similar characteristics but also over time. Age is thus unique from other elite characteristics in its effect on the time horizons of politicians. Future studies should investigate the extent to which age can similarly affect the decision making of politicians on other issues that involve trade-offs between short-term costs and long-term social returns such as climate change, government debt, and trade protectionism. Similarly, researchers should explore how the age of elected officials interacts with institutions that affect their time horizons, such as term limits.

In drawing broader lessons for policy making and existing theories, it is important to address the generalizability of these findings. Concerns about generalizability are common to RDDs, which have high internal validity but can struggle with external validity. The advantage of my design is that I can estimate the causal effect of electing a younger mayor on welfare expenditures among municipalities that otherwise should be very similar in observable and unobservable characteristics. However, the disadvantage is that the RDD only estimates the local average treatment effect for a subset of close elections between younger and older

candidates.[51] In the Appendix, I test whether similar patterns emerge if I instead analyze the full sample of elections using fixed effects regression (Table A12). Similar to the RD results, I find that mayors under 50 increase spending on child welfare by over 7% but do not have a significant effect on expenditures for elderly welfare. While these estimates are not causally identified, they corroborate the external validity of the RD findings by suggesting that younger mayors in settings beyond close elections between younger and older candidates can have a significant, if smaller impact on social spending for young families.

With that being said, more research is needed to explore the extent to which the results are generalizable beyond Japan. Japan is a rapidly aging society where social welfare is an especially salient issue and politicians tend to be much older than their average constituent. Like Japan, South Korea has one of the fastest-aging populations in the world and few young people in public office, while the United States also has relatively low youth representation despite its slower pace of demographic aging. A question for future studies is whether we see similar age differences among elected officials in their welfare policies in these settings, and how this compares to politicians in countries with much younger populations, such as Brazil and India, or countries where younger people have much greater descriptive representation in political institutions, such as Sweden and Denmark.

Relatedly, this study focuses on a setting where age is one of the most salient sources of identity-based variation, as Japanese mayors are overwhelmingly male and ethnically Japanese. This makes it easier to isolate the effect of a politician's age on policy outcomes, but it also highlights the need for future research on how age interacts with other social identities. For example, many studies find that female politicians are more likely to prioritize childcare and family policies than male politicians. Exploring whether the effects of electing younger politicians are amplified, diminished, or transformed when the representative is also a woman—or belongs to another underrepresented group—would offer important insights

[51] Notably, prior research has suggested that the uncertainty of competitive elections should shorten the expected time horizons of politicians and discourage investment. Finding a substantial effect of a mayor's age on child welfare investment following close elections in Japan, an arguably demanding test, may indicate that age differences among politicians will translate to other settings.

into how intersecting identities shape policy outcomes.

Lastly, researchers should investigate how age shapes elite behavior across other political offices and policy areas. While I focus on mayors to better estimate the causal effect of a constituency electing a younger representative on social spending, other studies could test whether younger legislators are more likely than their older colleagues to serve on committees that oversee childcare issues or respond to requests for welfare services from younger citizens. Beyond welfare, we know from other research that individuals of different ages often have different views on a wide range of issues, including same-sex marriage, immigration, gender equality, global governance, and environmental protection (Munger 2022; Wattenberg 2007). Future work should explore the extent to which younger politicians adopt different positions on these issues in office than older politicians.

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Does Youth Representation Matter for Social Spending?

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Online Appendix

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1 Japan Municipal Elections Dataset

1.1 Data Collection

My goal in creating the Japan Municipal Elections Dataset (JMED) was to collect data on all mayoral candidates and elections from 2006 to 2019 in Japan's 1,741 municipalities, which include 791 cities, 744 towns, 183 villages, and 23 special wards in Tokyo. As noted in the main text, I start from 2006 to avoid some of the challenges with handling the large wave of municipal mergers in the late 1990s and early 2000s (addressed in the next section), and end in 2019 to avoid the complications that the COVID-19 pandemic had for social spending.

The data collection process occurred in three stages. First, I used web scraping to construct the initial dataset by collecting information from a variety of online platforms in Japan that contain information on local candidates and elections, with the two main sources being Election Dot-Com (*Senkyo dotto komu*, https://go2senkyo.com) and Political Mountain (*Seijiyama*, https://seijiyama.jp). Second, together with a team of research assistants, we supplemented this initial information by looking through newspaper archives, municipal government pages, and individual candidate websites to fill in missing information. Finally, for the sample of elections included in the RDD (discussed later), we used these same online sources to collect additional information on candidates, including their education, family structure, party support, and prior experience in politics or government.

Once assembled, we took various steps to verify the validity of JMED using available data. First, given that Japan has fixed four-year terms between mayoral elections, we could use this fact to check that we were not missing any obvious elections. Second, Japan's Ministry of Internal Affairs and Communications does publish data on some mayoral candidates and elections, particularly for larger cities, which we could use to confirm that we were not missing any candidates in at least these races. Finally, we randomly selected 100 elections to check for errors and missing candidates. Although we did find some errors with respect to partisanship, we did not find any missing candidates. This gives us further confidence that

the dataset is providing a comprehensive accounting of candidates during this period.

In sum, the final JMED includes the names, ages, gender, incumbency, partisanship, and vote totals for 12,191 mayoral candidates in 6,215 elections from 2006 to 2019.

1.2 Handling Municipal Mergers

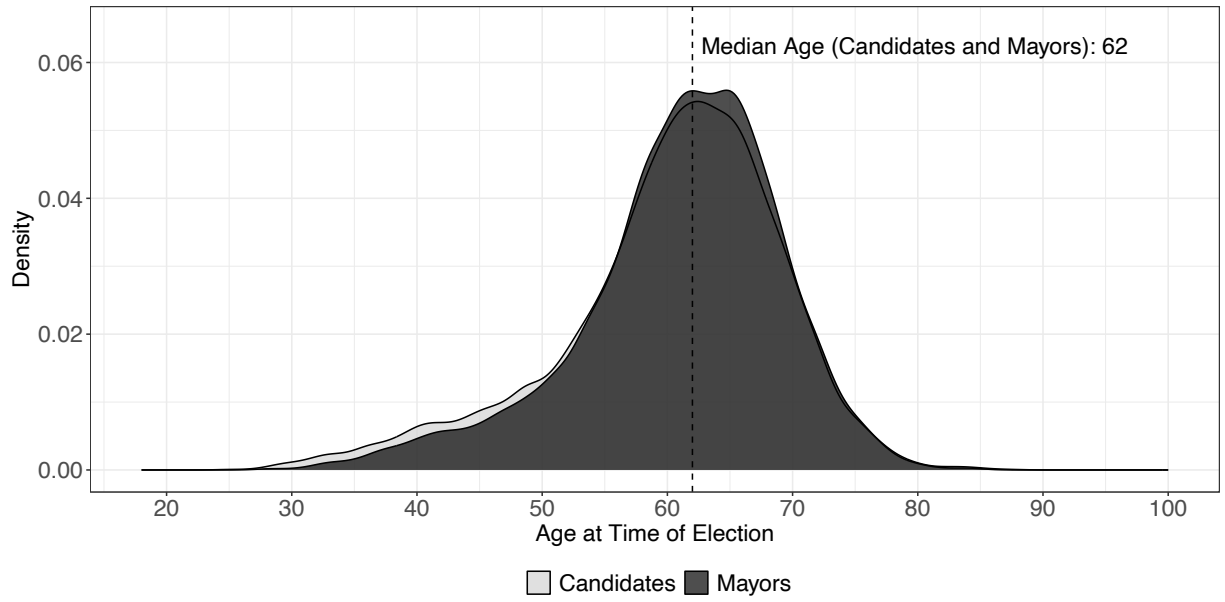
One complicating factor in constructing JMED was how to handle the phenomenon of municipal mergers—that is, the fact that the number of municipalities has not remained constant over time. As noted above and in the main text, beginning in the late 1990s, Japan experienced a large wave of municipal mergers that reduced the number of municipalities (excluding Tokyo special wards) from 3,232 in 1997 to 1,718 by 2016. The choice to begin the analysis with 2006 was made to avoid the complication that the large bulk of these mergers took place before the end of 2005, which would have complicated efforts to link the age of mayors to changes in social spending. By the start of 2006, nearly 80% of these mergers had already concluded (reducing the number of municipalities to 2,052) and over 90% had finished by the start of 2007 (reducing the number of municipalities to 1,816).

JMED thus excludes any elections that may have taken place in the 334 municipalities (most of them small towns and villages) that ceased to exist as independent government entities during this period. While most of these municipalities became part of larger towns and cities, the dataset also includes eight new cities that were established as a result of mergers. In total, less than 10% of JMED elections occurred in a municipality that experienced a merger between 2006 and 2016, with most mergers taking place in 2006 or 2007.

Later in this Appendix, I include some robustness checks to address the possibility that municipal mergers could be affecting the main results. Table A6 shows that all of the main results still hold when excluding any municipalities that experienced a merger during this period. Additionally, Table A7 shows that there is no discontinuity in having experienced a merger at the RDD threshold, which suggests municipalities that narrowly elect younger and older mayors do not differ substantially in past incidences of mergers.

1.3 Age Distribution of Candidates and Mayors

Figure A1: Age Distribution of Candidates and Mayors (2006–2019)



1.4 Trends Over Time

Figure A2: Age Distribution of Mayors (2006 vs. 2018)

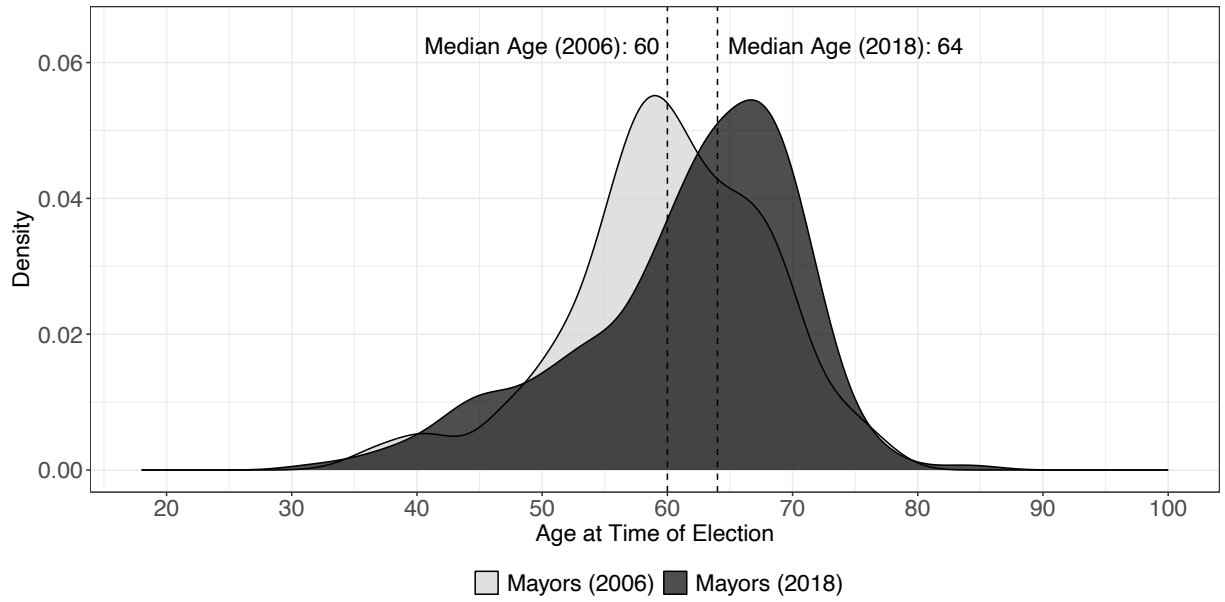
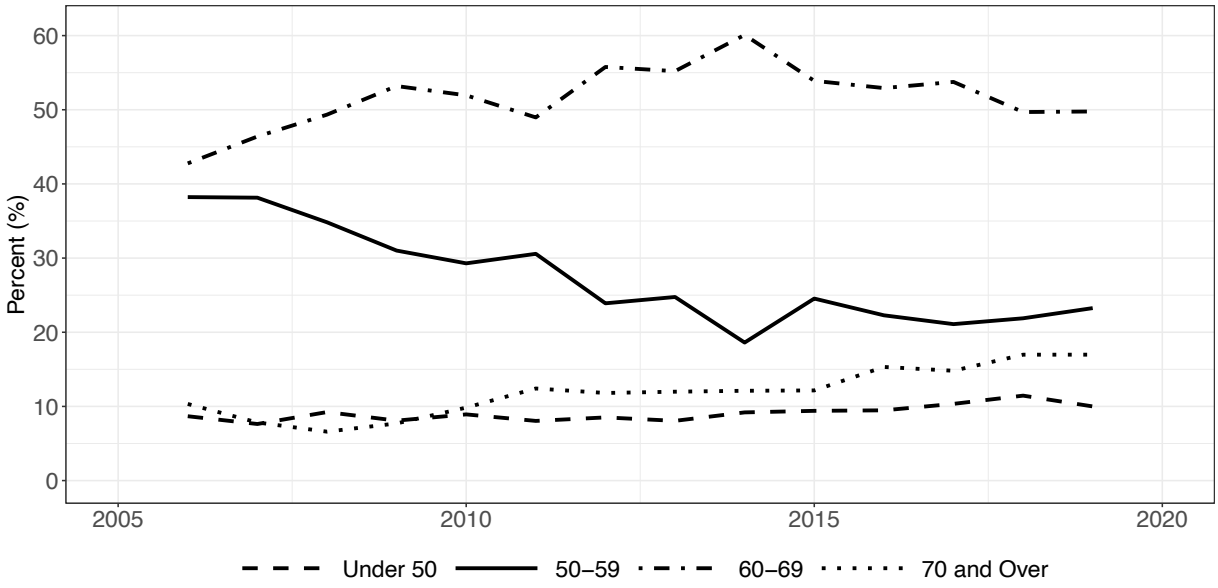


Figure A3: Mayors by Age Group (2006–2019)



2 Municipal Social Welfare

2.1 Local Public Finance Conditions Survey

The municipal budget data used in this paper comes from the Local Public Finance Conditions Survey (*chihō zaimu jōkyō chōsa*), an official dataset compiled annually by Japan's Ministry of Internal Affairs and Communications. The survey provides detailed data on the settled accounts of all Japanese municipalities, offering a centralized and standardized view into how local governments implement their budgets. The results are published annually as statistical tables and are also incorporated into the White Paper on Local Public Finance, which is submitted to the National Diet and plays an important role in fiscal analysis and in assessing municipal financial health. I downloaded the survey data from e-Stat (https://www.e-stat.go.jp/), Japan's official portal site for government statistics.

2.2 Mandatory vs. Discretionary Spending

In Japan's municipal budget system, expenditures are classified according to their economic nature into mandatory and discretionary categories. Mandatory spending refers to expenses that municipal governments are legally or structurally obligated to pay and which are generally difficult to reduce at the mayor's discretion. These include personnel costs such as salaries and pensions for public employees and elected officials; social assistance expenses, including centrally mandated entitlement programs such as the child allowance; and debt service obligations like bond redemption and interest payments.

Discretionary spending, by contrast, refers to expenditures that are allocated annually through the municipal budget process and over which the mayor has greater influence. These include investment expenses, such as the construction of new welfare facilities; subsidies, such as financial assistance to residents or local service providers; and other categories like goods, reserves, and transfers to other accounts.

In the main analysis, I focus on discretionary spending within the child and elderly welfare categories, specifically the three components that plausibly translate into visible or concrete welfare benefits for residents: Subsidies, Investment, and Goods. I exclude Reserves, Transfers, and Other Expenditures, which are part of discretionary spending but do not directly support welfare service delivery or resident-facing benefits.

2.3 Child vs. Elderly Welfare

A key benefit of the Local Public Finance Conditions Survey for this paper is that it distinguishes between spending on young families with children (child welfare) compared to older people (elderly welfare). Child welfare spending includes parental leave benefits, support for daycare and maternity centers, subsidies for families with children, and programs for children with disabilities. In contrast, elderly welfare spending includes public expenditures on senior care services, subsidies, and the construction or operation of facilities for the elderly.

Although Japan's welfare system overall is tilted toward the elderly (Lynch 2006), munic-

ipalities on average spend more per capita on child welfare than on elderly welfare, though there is considerable heterogeneity across jurisdictions. According to the 2020 White Paper on Local Finance, public welfare accounted for the largest category of municipal expenditures (36.3%), followed by general administration (12.2%), education (12.1%), and civil engineering (11.2%). Within total welfare spending, the largest component was child welfare (38.7%), followed by general social welfare (25.0%), elderly welfare (18.2%), and public assistance (17.8%).

2.4 Short-Term Spending vs. Long-Term Investment

To test the time horizons mechanism (H3), I further distinguish between short-term and long-term social spending. I define short-term spending as expenditures likely to yield immediate political returns, and long-term investment as expenditures that are costly in the short term but may produce sustained social benefits over time.

In the main analyses, I operationalize short-term spending using the Subsidies category, which includes discretionary transfers to individuals and organizations—for example, subsidies that reduce the cost of accessing childcare or elderly services. In contrast, I use the Investment category to capture long-term commitments, as these funds are used to construct, expand, or upgrade welfare facilities such as daycare centers and nursing homes.

While I selected these two categories because they offer the clearest conceptual fit with my theory, other spending categories could plausibly serve as alternatives for short-term spending. One possibility is Goods, which might appear more immediate or service-oriented. However, this category largely covers recurring operational costs such as consumable supplies, utilities, equipment rentals, and minor maintenance. While necessary for the delivery of welfare services, these expenditures do not align clearly with either side of the time horizons framework. They are less flexible and constituent-facing than subsidies, yet lack the delayed, capital-intensive nature of investment. As such, I do not treat them as either short- or long-term in the main analysis. Another potential candidate is Social Assistance, but this

falls under mandatory spending and primarily reflects national policy formulas rather than mayoral discretion.

Table A9, presented later in this Appendix, reports results for each of these additional categories to illustrate how the effects of electing a young mayor vary depending on how short-term spending is defined. I find no evidence that electing a mayor under 50 affects Social Assistance, while there is suggestive evidence ($p<0.10$) of a modest increase in spending on Goods. However, given the large and statistically significant effect on Investment, I interpret the results as consistent with the time horizons mechanism (H3): that younger mayors are especially likely to increase spending geared toward long-term investment.

3 Regression Discontinuity Design

3.1 Defining the RDD Sample

The regression discontinuity design (RDD) focuses on mayoral elections in which one of the top-two candidates was under 50 years old and the other was 50 or older. This produces a sample of 823 elections held in 602 municipalities across all 47 prefectures. Elections in which both candidates were under 50 or both were 50 or older are not included. Similarly, mayoral races in which only one candidate ran unopposed are excluded.

Following best practices, as outlined by Cattaneo, Idrobo and Titiunik (2019), I use a data-driven approach to define what counts as a "close election" by selecting the optimal bandwidth for each analysis to minimize mean square error. While the full RDD sample includes all 823 elections between younger and older candidates, only observations within the optimal bandwidth contribute to the estimation of the local average treatment effect (LATE) for each analysis. This data-driven definition of a "close election" varies by outcome, ranging from a vote-share bandwidth of 0.059 (n = 261 elections) to 0.137 (n = 553 elections). As discussed in the main text, I also test the robustness of each result to widening the threshold to include more elections.

To help readers better understand how the RDD sample compares to the broader pop-

ulation of mayoral elections, I present descriptive statistics for three groups in Table A1: (1) all JMED mayoral elections, (2) the full RDD sample (all younger vs. older candidate match-ups), and (3) the RDD sample with a bandwidth set to 0.10 (n = 461 elections).

3.2 Modeling Age as a Binary Treatment

The decision to model age as a binary variable—defining "young" mayors as those under 50—is grounded in both theoretical and empirical considerations. Theoretically, the central research question of this manuscript is not whether slightly younger mayors behave differently than slightly older ones, but whether electing a mayor who is categorically young, in terms of their position along the life span (Figure 1), leads to different policy decisions. The focus is on the consequences of youth underrepresentation. Similarly, all three mechanisms—life stage, time horizons, and electoral incentives stemming from proximity in age to younger voters—are derived from the decision to focus on the intrinsic and extrinsic motivations that might drive young politicians to represent the interests of young people in office.

Empirically, the distribution of candidates' ages supports a binary approach. As shown in Figure A4, mayors in Japan are relatively normally distributed around a median age of 62. Analyzing the effect of age as a continuous variable would therefore be driven primarily by comparisons among middle-aged and older candidates. Once the sample is restricted to elections between one candidate under 50 and one 50 or older, however, the distribution becomes bimodal, further justifying the use of a sharp cutoff.

One potential concern with the binary treatment is that observed differences might be driven by candidates who are only marginally different in age—for example, a 49-year-old versus a 50-year-old. However, this is extremely rare. In fact, there are no elections in the RDD sample where the top-two candidates differ in age by a single year. Across the 823 elections in this sample, the median age gap between candidates is 20 years. Only 13 elections (1.6%) feature a candidate age gap of less than 5 years, and just 84 (10.2%) have a gap smaller than 10 years. Table A5 shows that the results are robust to dropping any races

in which the top two candidates are within 10 years of each other. Additionally, Figure A6 shows the results hold for alternative choices of an age cutoff for a "young" mayor.

3.3 RDD Sample vs. All Elections

Figure A4: Age Distribution of Mayors (RDD Sample vs. All Elections)

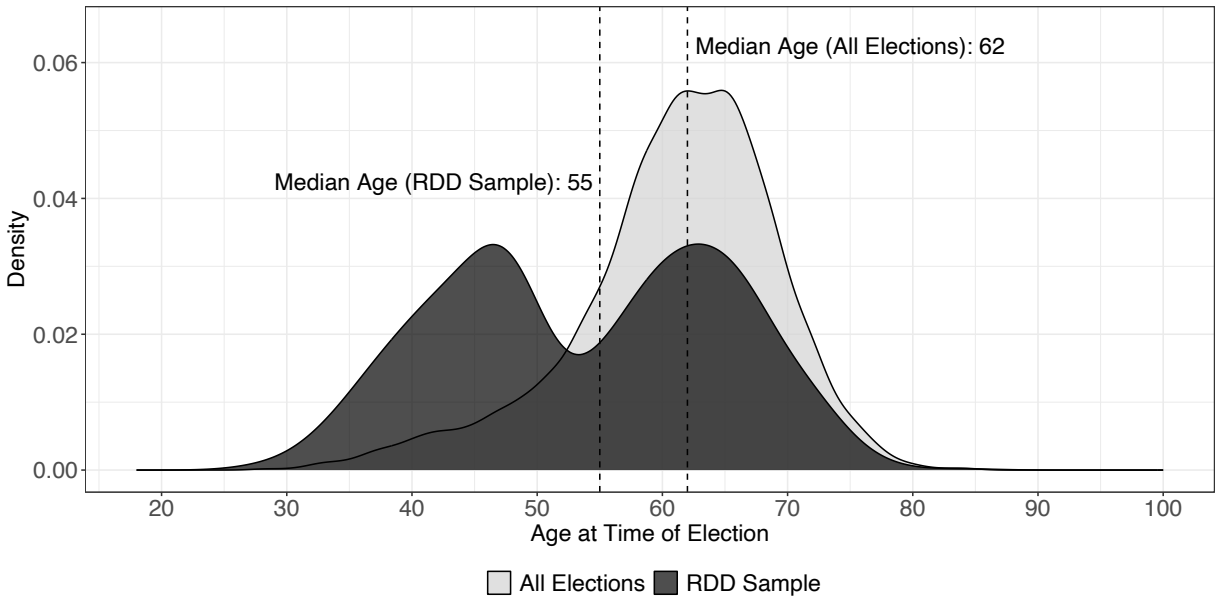


Table A1: RDD Sample vs. All Elections

	RDD Sample					
	Bandwidth = 0.1		Full Sample		All Elections	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
Population						
Total (thousands)	147.631	(319.395)	157.902	(322.898)	74.354	(191.940)
% Under 15	0.130	(0.022)	0.128	(0.022)	0.124	(0.024)
% 15 to 64	0.607	(0.049)	0.608	(0.051)	0.590	(0.055)
% 65 and Over	0.607	(0.049)	0.608	(0.051)	0.590	(0.055)
Region						
Hokkaido	0.059	(0.235)	0.051	(0.220)	0.104	(0.305)
Tohoku	0.061	(0.239)	0.062	(0.241)	0.131	(0.338)
Kanto	0.256	(0.437)	0.279	(0.449)	0.184	(0.388)
Chubu	0.176	(0.381)	0.182	(0.386)	0.178	(0.383)
Kansai	0.197	(0.398)	0.180	(0.384)	0.130	(0.336)
Chugoku	0.052	(0.222)	0.047	(0.213)	0.060	(0.237)
Shikoku	0.033	(0.178)	0.045	(0.207)	0.055	(0.227)
Kyushu	0.167	(0.373)	0.153	(0.360)	0.158	(0.365)
Spending (per capita)						
Child Welfare	3.928	(0.669)	3.970	(0.682)	67.249	(101.833)
Short-Term Spending	2.366	(0.882)	2.443	(0.896)	12.612	(24.851)
Long-Term Investment	1.970	(1.338)	1.993	(1.357)	18.606	(80.359)
Elderly Welfare	2.813	(1.012)	2.858	(1.032)	33.303	(61.291)
Infrastructure (per thousand)						
Daycare Centers	5.869	(4.163)	5.922	(4.341)	7.938	(9.340)
Elderly Care Centers	0.270	(0.221)	0.259	(0.204)	0.323	(0.300)
Elections	461		823		6,215	
Municipalities	390		602		1,741	
Prefectures	47		47		47	

Notes: JMED; Ministry of Internal Affairs and Communications (2020). Social spending and infrastructure are lagged one year to capture the year before the election. For social spending, child welfare spending is divided by the population under 15 and elderly welfare by the population 65 and over. For infrastructure, daycare centers represent the number per 1,000 children under age 5, while elderly care centers represent the number per 1,000 adults age 65 and over.*p<.1; **p<.05; ***p<.01.

Figure A5: Municipalities in the RDD Sample



Notes: Municipalities included in the RDD sample are shown in grey (see Table A1). Sources: JMED; Database of Global Administrative Areas (2018).

3.4 Summary Statistics

Table A2: RDD Summary Statistics

	Mean	SD	Min	Max	N
Candidate Under 50's Vote Margin	−0.016	(0.149)	−0.424	0.464	823
Δ Child Welfare Spending	0.137	(0.526)	−2.070	3.188	769
Δ Short-Term Spending	0.132	(0.556)	−4.917	2.621	769
Δ Long-Term Investment	0.169	(1.720)	−6.229	6.115	769
Δ Elderly Welfare Spending	−0.048	(0.466)	−2.588	2.262	769
Δ Short-Term Spending	−0.041	(0.403)	−2.961	3.076	769
Δ Long-Term Investment	0.016	(1.026)	−3.959	4.257	769

Notes: Mayoral elections where one candidate is under 50 years old. Variables report the change in logged per capita spending on child and elderly welfare from the year before to the year after the election. Elections held in 2019 are excluded because 2020 spending data are not used (see main text), reducing the sample size from 823 to 769.

3.5 Robustness Checks

I conduct several analyses to assess the sensitivity of the main findings. The RD results are robust to alternative modeling choices such as using higher-order polynomials, different bandwidths, and including control variables and year fixed effects (Tables A3 and A4). As noted earlier, I also find similar effects using different age cutoffs (Figure A6), focusing only on candidates with at least a ten-year age gap (Table A5), and excluding municipalities that experienced any mergers during this period (Table A6). Additionally, as a falsification test for the as-if random assumption of the RDD, I show that municipalities on either side of the election threshold are balanced in terms of their pre-existing population size, age demographics, and social welfare infrastructure and expenditures (Table A7). The one exception is that the result for the Chubu region is negative and significant. While this could be due to random chance, it is worth noting that the results are substantively similar if Chubu elections are excluded. A McCrary (2008) density test further indicates that there is no evidence of sorting near the threshold, which suggests that younger and older candidates do not differ in their ability to manipulate their vote share in close elections (Figure A7).

Table A3: Effect of Electing a Mayor Under 50 on Social Spending

DV	RD Estimate	SE	Robust CI	Specification	BW	N
Δ Child Welfare	0.356***	(0.134)	[0.094, 0.707]	Local Linear	0.078 (h)	340
Δ Child Welfare	0.273***	(0.091)	[0.130, 0.686]	Local Linear	0.156 ($2h$)	564
Δ Child Welfare	0.396***	(0.137)	[0.112, 0.728]	Quadratic	0.165 (h)	582
Δ Child Welfare	0.463**	(0.189)	[0.065, 0.921]	Cubic	0.190 (h)	624
Δ Elderly Welfare	0.051	(0.098)	[–0.198, 0.248]	Local Linear	0.085 (h)	367
Δ Elderly Welfare	0.107	(0.076)	[–0.153, 0.260]	Local Linear	0.168 ($2h$)	587
Δ Elderly Welfare	−0.034	(0.140)	[–0.379, 0.238]	Quadratic	0.101 (h)	436
Δ Elderly Welfare	−0.109	(0.171)	[–0.515, 0.240]	Cubic	0.132 (h)	511

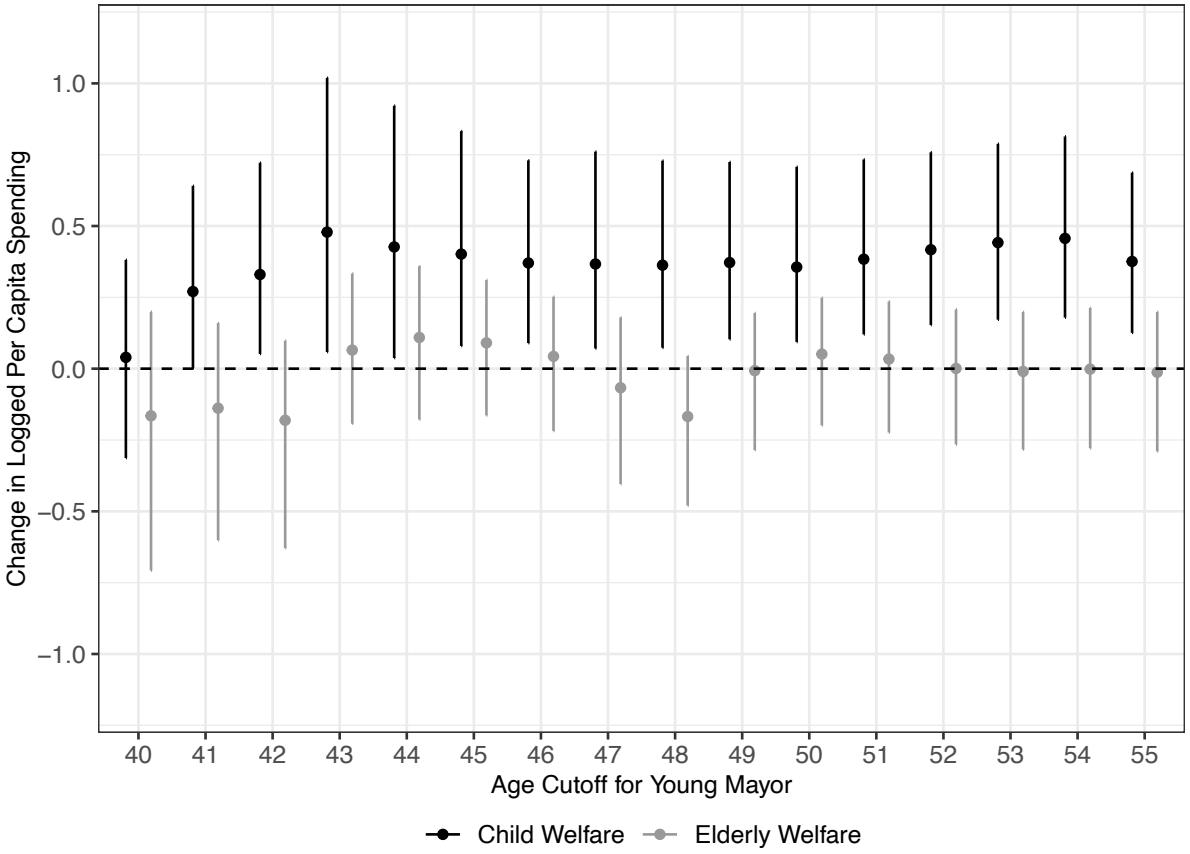
Notes: Table reports regression discontinuity (RD) estimates, conventional standard errors (SE), and robust 95% confidence intervals (Robust CI), with errors clustered by municipality. For bandwidths (BW), h represents the optimal bandwidth chosen to minimize mean square error. *p<.1; **p<.05; ***p<.01.

Table A4: Effect of Electing a Mayor Under 50 on Social Spending (With Controls)

DV	RD Estimate	SE	Robust CI	Specification	BW	N
Δ Child Welfare	0.304**	(0.135)	[0.035, 0.652]	Local Linear	0.078 (h)	335
Δ Child Welfare	0.251***	(0.091)	[0.100, 0.658]	Local Linear	0.154 ($2h$)	562
Δ Child Welfare	0.370***	(0.135)	[0.090, 0.695]	Quadratic	0.168 (h)	587
Δ Child Welfare	0.434**	(0.187)	[0.037, 0.887]	Cubic	0.191 (h)	627
Δ Elderly Welfare	−0.023	(0.101)	[–0.284, 0.174]	Local Linear	0.080 (h)	345
Δ Elderly Welfare	0.083	(0.078)	[–0.202, 0.221]	Local Linear	0.158 ($2h$)	567
Δ Elderly Welfare	−0.091	(0.142)	[–0.440, 0.184]	Quadratic	0.099 (h)	423
Δ Elderly Welfare	−0.146	(0.170)	[–0.548, 0.204]	Cubic	0.133 (h)	511

Notes: Controls include incumbency, gender, population, and year fixed effects. Table reports regression discontinuity (RD) estimates, conventional standard errors (SE), and robust 95% confidence intervals (Robust CI), with errors clustered by municipality. For bandwidths (BW), h represents the optimal bandwidth chosen to minimize mean square error. *p<.1; **p<.05; ***p<.01.

Figure A6: Effect of Young Mayor on Social Spending Across Age Cutoffs



Notes: The figure presents the RD estimates of electing a mayor under a given age cutoff on the change in logged per capita spending on child welfare (black) and elderly welfare (grey).

Table A5: Effect of Electing a Mayor Under 50 on Social Spending (10+ Year Age Gap)

DV	RD Estimate	SE	Robust CI	Specification	BW	N
Δ Child Welfare	0.352***	(0.113)	[0.127, 0.644]	Local Linear	0.079 (h)	341
Δ Child Welfare	0.288***	(0.082)	[0.156, 0.626]	Local Linear	0.158 ($2h$)	567
Δ Child Welfare	0.388***	(0.138)	[0.092, 0.714]	Quadratic	0.126 (h)	500
Δ Child Welfare	0.390**	(0.185)	[–0.040, 0.800]	Cubic	0.148 (h)	549
Δ Elderly Welfare	−0.051	(0.112)	[–0.339, 0.161]	Local Linear	0.071 (h)	304
Δ Elderly Welfare	0.053	(0.084)	[–0.306, 0.156]	Local Linear	0.142 ($2h$)	533
Δ Elderly Welfare	−0.130	(0.145)	[–0.489, 0.149]	Quadratic	0.098 (h)	422
Δ Elderly Welfare	−0.198	(0.175)	[–0.609, 0.161]	Cubic	0.131 (h)	509

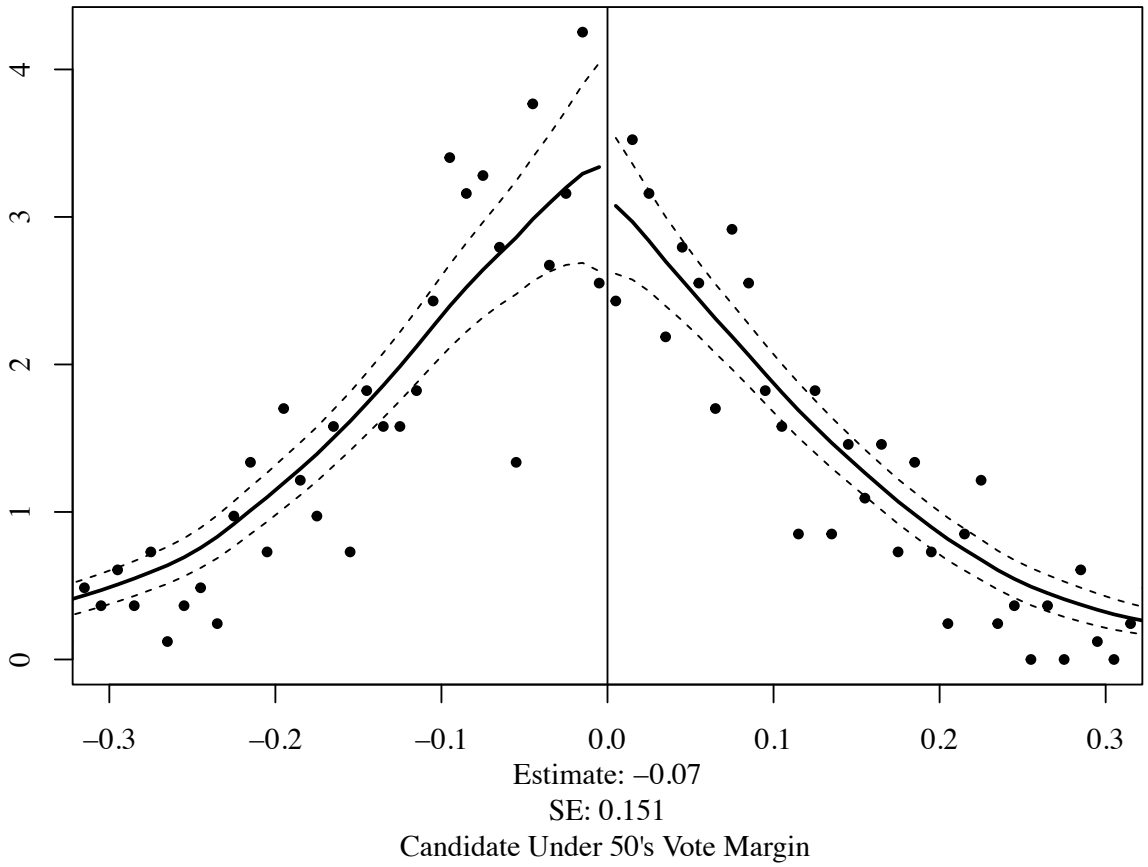
Notes: Sample limited to candidates who are at least 10 years apart in age. Table reports regression discontinuity (RD) estimates, conventional standard errors (SE), and robust 95% confidence intervals (Robust CI), with errors clustered by municipality. For bandwidths (BW), h represents the optimal bandwidth chosen to minimize mean square error. *p<.1; **p<.05; ***p<.01.

Table A6: Effect of Electing a Mayor Under 50 on Social Spending (Excluding Mergers)

DV	RD Estimate	SE	Robust CI	Specification	BW	N
Δ Child Welfare	0.306**	(0.133)	[0.048, 0.653]	Local Linear	0.080 (h)	345
Δ Child Welfare	0.226**	(0.092)	[0.082, 0.634]	Local Linear	0.160 ($2h$)	568
Δ Child Welfare	0.360**	(0.142)	[0.068, 0.704]	Quadratic	0.158 (h)	567
Δ Child Welfare	0.429**	(0.209)	[–0.005, 0.934]	Cubic	0.171 (h)	590
Δ Elderly Welfare	0.048	(0.105)	[–0.220, 0.258]	Local Linear	0.090 (h)	387
Δ Elderly Welfare	0.100	(0.082)	[–0.168, 0.276]	Local Linear	0.180 ($2h$)	603
Δ Elderly Welfare	−0.054	(0.151)	[–0.426, 0.238]	Quadratic	0.104 (h)	449
Δ Elderly Welfare	−0.156	(0.195)	[–0.623, 0.236]	Cubic	0.125 (h)	499

Notes: Sample limited to municipalities that did not experience any mergers from 2006 to 2019. Table reports regression discontinuity (RD) estimates, conventional standard errors (SE), and robust 95% confidence intervals (Robust CI), with errors clustered by municipality. For bandwidths (BW), h represents the optimal bandwidth chosen to minimize mean square error. *p<.1; **p<.05; ***p<.01.

Figure A7: McCrary Density Test



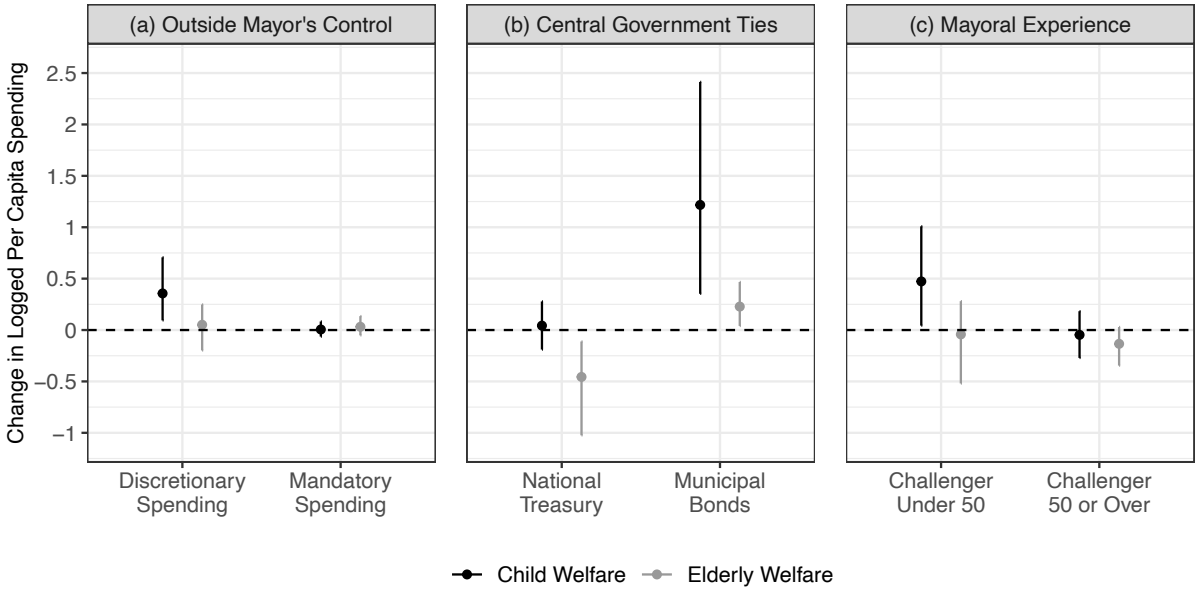
Notes: Plot shows the density of the candidate under 50's vote margin is continuous at the threshold.

Table A7: Balance Checks for Pre-Treatment Covariates

	RD Estimate	SE	Robust CI	BW (h)	N
Population					
Total (thousands)	−49.091	(71.783)	[–218.893, 134.862]	0.109	493
% Under 15	−0.006	(0.006)	[–0.021, 0.006]	0.069	316
% 15–64	−0.018	(0.012)	[–0.049, 0.006]	0.076	349
% 65 and Over	−0.018	(0.012)	[–0.049, 0.006]	0.076	349
Region					
Hokkaido	0.007	(0.059)	[–0.117, 0.163]	0.090	418
Tohoku	−0.051	(0.047)	[–0.154, 0.063]	0.129	541
Kanto	−0.026	(0.088)	[–0.241, 0.172]	0.080	372
Chubu	−0.197**	(0.098)	[–0.452, –0.006]	0.061	287
Kansai	0.042	(0.091)	[–0.174, 0.262]	0.092	432
Chugoku	0.067	(0.044)	[–0.024, 0.174]	0.119	514
Shikoku	−0.012	(0.045)	[–0.116, 0.096]	0.078	358
Kyushu	0.090	(0.076)	[–0.071, 0.277]	0.104	483
Municipal Merger					
Prior Merger	0.033	(0.057)	[–0.095, 0.163]	0.137	553
Spending (per capita)					
Child Welfare	−0.226	(0.152)	[–0.618, 0.084]	0.118	512
Short-Term Spending	0.124	(0.219)	[–0.326, 0.690]	0.081	376
Long-Term Investment	−0.486	(0.324)	[–1.331, 0.192]	0.091	430
Elderly Welfare	0.092	(0.175)	[–0.305, 0.498]	0.127	538
Infrastructure (per thousand)					
Daycare Centers	−0.251	(1.410)	[–3.598, 3.212]	0.082	274
Elderly Care Centers	−0.022	(0.054)	[–0.163, 0.088]	0.065	234

Notes: Table reports regression discontinuity (RD) estimates, conventional standard errors (SE), and robust 95% confidence intervals (Robust CI), with errors clustered by municipality. All models use local linear regression and optimal bandwidths (h) chosen to minimize mean square error.*p<.1; **p<.05; ***p<.01.

Figure A8: Ruling Out Alternative Explanations



Notes: The figure presents the RD estimates of electing a mayor under 50 (panels a and b) or a challenger candidate (panel c) on the change in logged per capita spending (or revenue) for child welfare (black) and elderly welfare (grey) across three panels. Panel (a) compares discretionary and mandatory spending. Panel (b) compares revenues from national treasury disbursements with that from municipal bonds. Panel (c) estimates the effect of electing a challenger candidate using a new sample of close elections between challengers and incumbents, which has been divided by whether the challenger is under 50.

3.6 Ruling Out Alternative Explanations

In addition to robustness checks, I seek to rule out three potential alternative explanations. Figure A8 shows the results of these tests, using a similar presentational style as Figure 4.

The first possibility is that the narrow election of a younger mayor could coincide with social changes outside of the mayor's control that affect welfare spending. To address this potential threat, I use placebo tests to show that younger mayors only affect changes in discretionary spending and not mandatory spending (panel a), the latter of which is predominantly determined by the central government according to formulas based on municipal income and demographics. The lack of any significant effects for mandatory spending suggest that my findings capture the discretion of younger mayors as opposed to some coinciding change in national welfare policy or the socioeconomic characteristics of municipalities.

Another possible explanation is that younger and older mayors may differ in their ties to

the central government, which represent a key revenue source for municipalities (Catalinac, Bueno de Mesquita and Smith 2020) and could affect each mayor's ability to fund welfare programs. However, in panel (b), I do not find that mayors under 50 receive significantly more disbursements from the national treasury than mayors 50 or older for child welfare, though they do receive significantly less for elderly welfare.[1] Instead, younger mayors support their increased spending and investment in child welfare by reallocating money within the budget and issuing municipal bonds to fund infrastructure development, the latter of which I highlight in panel (b) (see also Table A11).[2]

Finally, the last potential threat relates to the recognition that age, like other politician characteristics, is a bundled treatment (Bertoli and Hazlett 2025; Marshall 2022). Mayors of different ages may also share other characteristics in common, and it could be these factors rather than age that drive the observed effects on social spending. To address this threat, I again rely on RD tests to assess whether there are any discontinuities in mayoral-level covariates at the election threshold (Table A8). While younger and older mayors do differ in ways apart from their age, most differences are not significant. The one exception, as suggested by Figure 1, is that younger mayors are more likely to be entering office with less experience serving as mayor.

Is this difference in mayoral experience the main driver of the observed age differences in social spending? To assess this possibility, I first focus only on elections where the top-two candidates featured an incumbent facing off against a challenger candidate, and then estimate the RD effect of narrowly electing a challenger (i.e., new mayor) over an incumbent. In panel (c), I then split this sample by whether the challenger candidate is under 50 years old. I find that electing a challenger under 50 has a significant and positive effect on child welfare spending, whereas electing a challenger that is 50 or older has no similar effect. As

[1] More generally, I find no evidence that a mayor's age significantly affects other budget categories, whether expenditures or revenues (Table A10).

[2] The greater willingness to issue municipal bonds is notable because it suggests that younger mayors are also more likely to shift some of the costs of their child welfare policies toward the longer term.

Table A8: Other Characteristics and Experiences of Mayors Under 50

	RD Estimate	SE	Robust CI	BW (h)	N
Characteristics					
Female	−0.010	(0.035)	[–0.098, 0.062]	0.086	398
Education	0.135	(0.093)	[–0.087, 0.351]	0.104	270
Liberal Democratic Party	0.030	(0.051)	[–0.093, 0.147]	0.087	372
Any Political Party	−0.044	(0.063)	[–0.207, 0.080]	0.069	296
Experience					
Mayor	−0.582***	(0.084)	[–0.801, –0.426]	0.084	390
Municipal Assembly	0.219	(0.181)	[–0.200, 0.663]	0.076	176
Prefectural Assembly	0.039	(0.123)	[–0.221, 0.353]	0.090	218
Municipal Bureaucrat	0.007	(0.130)	[–0.276, 0.336]	0.083	177
National Bureaucrat	0.132	(0.095)	[–0.108, 0.346]	0.099	220

Notes: Education: 0=high school, 1=college, 2=graduate school. Liberal Democratic Party and Any Political Party indicates campaign support or recommendation from the respective party. Table reports regression discontinuity (RD) estimates, conventional standard errors (SE), and robust 95% confidence intervals (Robust CI), with errors clustered by municipality. All models use local linear regression and optimal bandwidths (h) chosen to minimize mean square error. *p<.1; **p<.05; ***p<.01.

a result, it is unlikely that the effect of age is purely a proxy for past mayoral experience.[3]

[3]Table A4 further shows that the main RD results hold when controlling for past mayoral experience.

Table A9: Effect of Electing a Mayor Under 50 on Child and Elderly Welfare Spending

	RD Estimate	SE	Robust CI	BW (h)	N
Child Welfare					
Δ Mandatory Spending	0.005	(0.031)	[–0.064, 0.083]	0.081	350
Δ Social Assistance	−0.005	(0.042)	[–0.102, 0.096]	0.078	334
Δ Personnel	0.009	(0.042)	[–0.066, 0.124]	0.059	261
Δ Discretionary Spending	0.356***	(0.134)	[0.094, 0.707]	0.078	340
Δ Subsidies	−0.202**	(0.103)	[–0.474, –0.017]	0.062	273
Δ Investment	0.884**	(0.434)	[0.040, 2.022]	0.079	342
Δ Goods	0.123*	(0.064)	[–0.005, 0.290]	0.102	437
Elderly Welfare					
Δ Mandatory Spending	0.032	(0.040)	[–0.052, 0.135]	0.099	423
Δ Social Assistance	0.066	(0.068)	[–0.102, 0.225]	0.098	422
Δ Personnel	−0.021	(0.075)	[–0.216, 0.129]	0.109	458
Δ Discretionary Spending	0.051	(0.098)	[–0.198, 0.248]	0.085	367
Δ Subsidies	−0.014	(0.078)	[–0.211, 0.146]	0.070	300
Δ Investment	0.172	(0.214)	[–0.343, 0.673]	0.104	448
Δ Goods	0.052	(0.051)	[–0.060, 0.181]	0.119	478

Notes: Main analyses focus on Discretionary Spending, Subsidies (proxy for Short-Term Spending), and Investment (proxy for Long-Term Investment). Table reports regression discontinuity (RD) estimates, conventional standard errors (SE), and robust 95% confidence intervals (Robust CI), with errors clustered by municipality. All models use local linear regression and optimal bandwidths (h) chosen to minimize mean square error. *p<.1; **p<.05; ***p<.01.

Table A10: Effect of Electing a Mayor Under 50 on Municipal Expenditures and Revenues

	RD Estimate	SE	Robust CI	BW (h)	N
Expenditures					
Δ General	−0.032	(0.063)	[–0.170, 0.127]	0.095	414
Δ Welfare	0.036*	(0.021)	[–0.010, 0.090]	0.101	436
Δ Child	0.356***	(0.134)	[0.094, 0.707]	0.078	340
Δ Elderly	0.051	(0.098)	[–0.198, 0.248]	0.085	367
Δ Social	−0.002	(0.030)	[–0.075, 0.064]	0.086	368
Δ Protection	0.037	(0.051)	[–0.071, 0.166]	0.089	378
Δ Disaster	0.057	(0.096)	[–0.146, 0.303]	0.079	341
Δ Sanitation	0.012	(0.057)	[–0.122, 0.148]	0.084	360
Δ Labor	0.049	(0.101)	[–0.181, 0.286]	0.107	453
Δ Agriculture	0.025	(0.125)	[–0.269, 0.303]	0.080	343
Δ Industry	−0.040	(0.116)	[–0.329, 0.196]	0.097	417
Δ Civil	0.097	(0.061)	[–0.019, 0.259]	0.062	270
Δ Fire	0.002	(0.045)	[–0.099, 0.110]	0.085	362
Δ Education	0.188*	(0.104)	[–0.008, 0.461]	0.064	278
Revenues					
Δ Local Allocation Tax	0.034	(0.097)	[–0.205, 0.269]	0.103	436
Δ Local Taxes	−0.014	(0.011)	[–0.037, 0.013]	0.096	413
Δ National Treasury Disbursements	0.021	(0.085)	[–0.178, 0.225]	0.083	355
Δ Prefectural Treasury Disbursements	0.089	(0.148)	[–0.277, 0.421]	0.077	329

Notes: Table reports regression discontinuity (RD) estimates, conventional standard errors (SE), and robust 95% confidence intervals (Robust CI), with errors clustered by municipality. All models use local linear regression and optimal bandwidths (h) chosen to minimize mean square error. *p<.1; **p<.05; ***p<.01.

Table A11: Effect of Electing a Mayor Under 50 on Revenues for Child Welfare

	RD Estimate	SE	Robust CI	BW (h)	N
Δ General Resources	0.046**	(0.022)	[0.003, 0.104]	0.067	291
Δ National Treasury Disbursements	0.043	(0.096)	[–0.190, 0.278]	0.087	372
Δ Prefectural Treasury Disbursements	0.039	(0.052)	[–0.078, 0.164]	0.097	421
Δ Local Bonds	1.217***	(0.454)	[0.353, 2.414]	0.060	266
Δ Other	−0.012	(0.109)	[–0.295, 0.231]	0.098	421

Notes: Table reports regression discontinuity (RD) estimates, conventional standard errors (SE), and robust 95% confidence intervals (Robust CI), with errors clustered by municipality. All models use local linear regression and optimal bandwidths (h) chosen to minimize mean square error. *p<.1; **p<.05; ***p<.01.

3.7 External Validity

Table A12: Effect of Electing a Mayor Under 50 on Social Spending (Fixed Effects)

	Estimate	SE	N
Δ Child Welfare	0.071**	(0.033)	5,685
Δ Elderly Welfare	0.032	(0.039)	5,685

Notes: Linear models with municipality and year fixed effects. Standard errors (SE) clustered by municipality in parentheses. *p<.1; **p<.05; ***p<.01.

4 Interviews with Mayors and Bureaucrats

4.1 Structure and Ethical Considerations

Semi-structured interviews were conducted under the University of California, San Diego's Human Research Protections Program's Protocol (Project 181763). The project was deemed exempt from full IRB review because the subjects are all public officials and the confidentiality of the interviewees will be maintained throughout the research and thereafter.

The interviewees were informed about the aims of the project, interview process, confidentiality, and informed consent procedures. All interviews were audio-recorded, conducted in the Japanese language, and lasted an average of 45 minutes. All participants were made aware during consent that they had the right to stop the recording or erase the recording at any time during the interview. The audio files were professionally transcribed by someone who signed a non-disclosure agreement. I then personally translated the selected quotes in this article into English.

To protect the confidentiality of my participants, all of the interview notes and recordings are kept in password-protected files on a private computer and labelled using date and unique identification numbers. There is no electronic record of interviewee names, and the key linking identification numbers to interviewee names is kept in my locked office. In the main text, I list interview dates but not locations to protect the anonymity of interviewees.

4.2 Verbal Consent Protocol

Hello, my name is Charles McClean, and I am a PhD Candidate in the Department of Political Science at the University of California, San Diego. I am the principal investigator conducting a research study to find out about the effect of age on the behavior and policy preferences of elected officials. You have been asked to participate in this study because you are either a mayor or municipal bureaucrat in Japan. There will be approximately 30 participants in this study.

If you agree to be in this study, you will engage in a one-time interview ranging from 30 to 60 minutes. The interview will be audio-recorded and transcribed. The audio-recording is optional, and you may still participate without being recorded. During the interview, we will discuss [your/your mayor's] policy priorities, campaign activities, leadership style, and connection with [your/their] constituents.

Research records will be kept confidential to the extent allowed by law. In my written reports, small excerpts from the interview may be quoted. I will make sure to use a pseudonym to keep your identity confidential in my written reports, digital recordings, and transcriptions.

Participation in this study is entirely voluntary. Some risks of being interviewed are that you may, at times, experience feelings of anxiety, fatigue, or embarrassment. I want to assure you that should you experience discomfort during this interview you can refuse to answer the questions or ask to stop/withdraw from the interview without any adverse consequences.

If you have any questions about this study, you may contact me through the information listed on this page. This page also includes the contact information for the Human Research Protections Program Office at my university, which provides oversight for this study.

Do you have any questions at this time? Do you agree to participate?

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